Proteomics-Based Patient Stratification, and Therapeutic Target Discovery in Pediatric Brain Tumours

CPTAC/CHOP Brain Cancer Cohort (brain_cptac_2020)

2026-04-29

Contents

Overview . . . 1
1. Packages & Configuration . . . 3
2. Data Loading . . . 3
3. Quality Control . . . 4
4. Exploratory Analysis . . . 6
5. Dimensionality Reduction (PCA to NMF input) . . . 14
6. Selecting the Number of Clusters (k) . . . 14
7. Consensus NMF Clustering . . . 17
8. Cluster Annotation . . . 18
9. Histological Composition per Cluster . . . 19
10. Survival Analysis (Kaplan–Meier) . . . 21
11. Cox Proportional Hazards Model . . . 22
12. Differential Protein Expression (limma one-vs-rest) . . . 24
13. Random Forest (classification by histology) . . . 25
14. Therapeutic Target Discovery . . . 25
15. Molecular Surrogate Classifier . . . 30
References . . . 32

Overview

This notebook reproduces and extends the proteomics-based patient stratification from Petralia et al. (Cell, 2020) using the publicly available CPTAC/CHOP pediatric brain cancer dataset (`brain_cptac_2020` on cBioPortal). Building on the unsupervised NMF subtypes, it asks two complementary precision-oncology questions:

1. **Therapeutic target discovery (§14).** For each proteomic subtype, can we propose a panel of candidate drug targets supported by multiple, independent lines of evidence, cluster-specific overexpression (limma), inferred kinase activity (KSEA on the phosphoproteome), brain-cancer cell-line dependency (DepMap CRISPR essentiality), and approved-drug coverage (OpenTargets / DGIdb)?
2. **Molecular surrogate classification (§15).** Can the proteome serve as a surrogate predictor of clinically actionable molecular markers (WHO grade, IDH/H3F3A driver status) that today require sequencing? With 100+ patients per outcome class this is the only branch where a properly held-out predictive claim is statistically credible in this cohort.

Dataset: 218 tumour samples from 199 patients across 7 histological subtypes (LGG, Ependymoma, HGG, Medulloblastoma, Ganglioglioma, Craniopharyngioma, ATRT), each with WGS, RNA-seq, global proteomics, and phosphoproteomics.

Data files used from `./brain_cptac_2020`

File	Used for
`data_clinical_patient.txt`	Demographics, OS (`os_status`, `os_months`), DFS (`dfs_status`, `dfs_months`), WHO grade, age at diagnosis, sex, and pre-computed driver-mutation flags (`braf_status`, `h3f3a_ctnnb1_status`, `ependymoma_rela_status`, `hgg_h3f3a_status`, `lgg_braf_status`, `ctnnb1_status`)
`data_clinical_sample.txt`	Sample-level metadata: `cancer_type_detailed`, `tumor_type`, treatment fields (`extent_of_tumor_resection`, `radiation`, `chemotherapy`), tissue site
`data_protein_quantification.txt`	Global proteomics matrix: feature space for clustering, differential expression, and the molecular-subtype classifier (§15)
`data_phosphoprotein_quantification.txt`	Phosphosite-level abundances: feeds per-cluster kinase-activity inference (KSEA) in §14

Risk

1. **Prognostic risk**: the descriptive question *"do these proteomic subtypes have different survival outcomes?"*, addressed in §10–11 with KM curves and a Cox model on cluster + extended clinical baseline. Because the clusters were defined unsupervised on the proteome (no survival input), this section is honest and free from outcome leakage. With only 36 OS events across 218 patients, however, this cohort is too small to support a credible per-patient predictive risk score (the rule of thumb is 10 events per parameter), so we deliberately do not attempt one.
2. **Therapeutic risk and opportunity**: the precision-oncology question *"what targets is each subtype most likely to depend on, and which of those are druggable?"*, addressed in §14, where each cluster is characterised by overexpressed proteins, inferred kinase activity from phosphoproteomics, brain cell-line dependency profiles, and approved-drug coverage. The deliverable is a per-cluster panel of candidate targets, not a per-patient score.

The §15 classifier sidesteps the OS-event-count problem entirely by predicting molecular outcomes (WHO grade and driver status) where the cohort has 100+ events per class, allowing a properly cross-validated predictive claim.

Pipeline summary

```
Local data files
      load + clean column names
    v
Quality control
      remove high-missing proteins, KNN imputation
    v
Exploratory analysis
      PCA, PLS-DA, sPLS-DA, UMAP
    v
Dimensionality reduction
      top 25 PCs, shift to non-negative for NMF
    v
k selection
      cophenetic / dispersion / silhouette across k = K_MIN..K_MAX
    v
```

```
Consensus NMF clustering
      N runs at chosen k, consensus matrix, final cluster labels
    v
Biological characterisation
       Histological composition per cluster
       Kaplan-Meier survival (log-rank test)
       Cox model, extended clinical baseline + cluster
       Differential protein expression (limma one-vs-rest)
       Variable importance (RF)
    v
Therapeutic target discovery (§14)
       Phospho one-vs-rest contrasts per cluster
       Kinase activity inference (KSEA, OmniPath substrates)
       DepMap brain-line dependency for cluster markers
       Druggability annotation (OpenTargets / DGIdb)
       Integrated per-cluster therapeutic panel
    v
Molecular surrogate classifier (§15)
       Proteomic predictor of WHO grade and driver status
         with held-out cross-validation (honest predictive claim)
```

---

1. Packages & Configuration

All libraries are loaded up front so that every chunk runs in a known environment.

```
## Configuration ready. Data directory: ./brain_cptac_2020
```

---

2. Data Loading

We read flat files downloaded from cBioPortal. All files live in `DATA_DIR`. Column names are cleaned with `janitor::clean_names()` so that everything downstream uses consistent `snake_case`.

```
## Files loaded.
```

```
##   Clinical patients: 199 rows
```

```
##   Clinical samples: 218 rows
```

```
##   Proteomics rows: 6429 (should equal n_proteins)
```

```
##   Phosphoproteomics rows: 4548 (phosphosites)
```

The proteomics file ships in protein × sample orientation. We transpose it to samples × proteins because every clustering / modelling library we use expects samples as rows. We also fix the rownames: R prepends `x` to numeric-starting column names and converts hyphens to underscores when reading column headers, so we reverse both transformations to recover the canonical `7316-NNNN` sample IDs.

```
## Proteomics matrix: 218 samples × 6429 proteins
```

```
## Phosphoproteomics matrix: 217 samples × 4548 phosphosites
```

```
## Patients with OS data: 192   events: 40
```

```
## Patients with DFS data: 192   events: 90
```

```
## Histological subtypes:
```

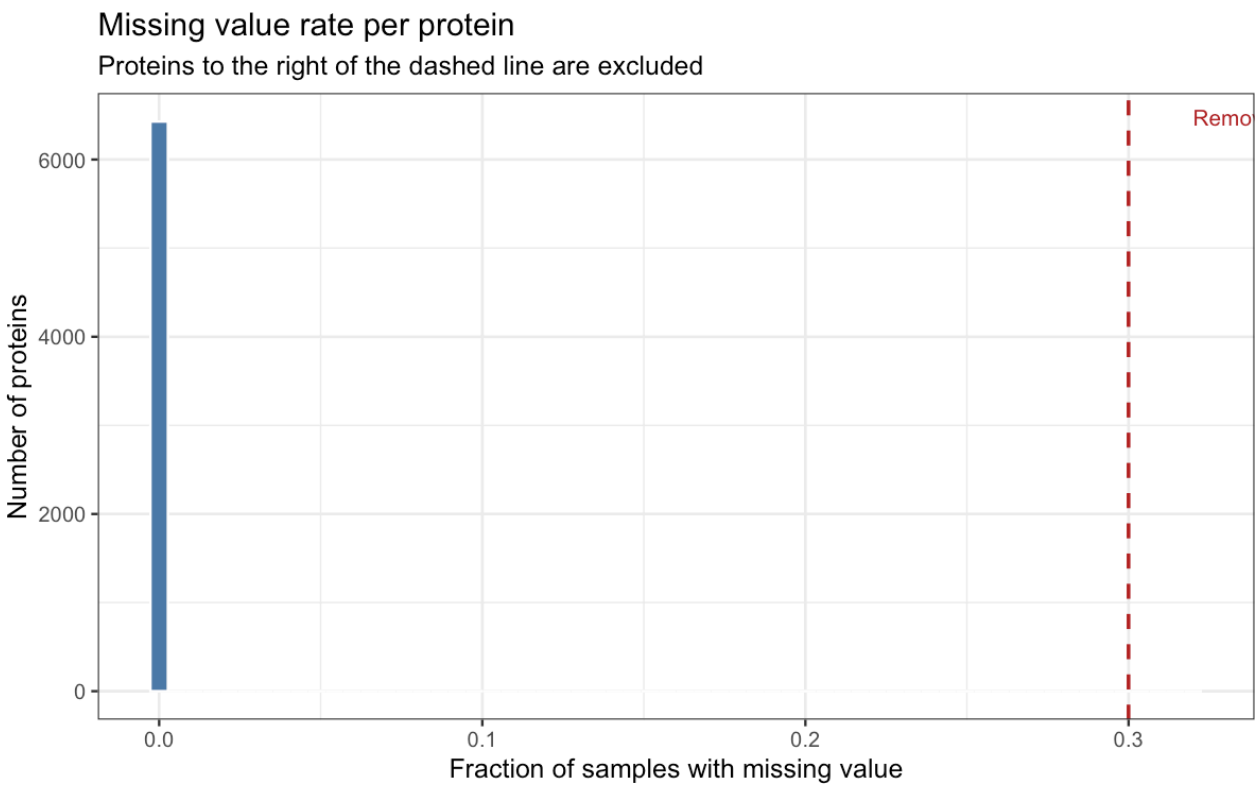
```
##                        cancer_type_detailed  n
## 1               Pediatric Low Grade Gliomas 93
## 2                         Ependymomal Tumor 32
## 3              Pediatric High Grade Gliomas 25
## 4                           Medulloblastoma 22
## 5                             Ganglioglioma 18
## 6 Craniopharyngioma, Adamantinomatous Type 16
## 7          Atypical Teratoid/Rhabdoid Tumor 12
```

---

3. Quality Control

Two systematic issues in mass-spectrometry proteomics must be addressed before clustering or modelling:

1. **Missing values.** Proteins below the detection limit are absent. Missingness is non-random (low-abundance proteins drop out more often). We remove proteins missing in more than `MISSING_THRESHOLD` of samples, then impute remaining NAs with KNN imputation.
2. **Outlier samples.** Samples missing > 40% of the remaining proteins likely reflect technical failures and should be dropped before any analysis that assumes a complete matrix.




```
## Proteins removed (> 30% missing): 0 | Remaining: 6429

## No outlier samples detected.

## Running KNN imputation (k = 10) on global proteome...

## Cluster size 6429 broken into 5848 581
## Cluster size 5848 broken into 3800 2048
## Cluster size 3800 broken into 678 3122
## Done cluster 678
## Cluster size 3122 broken into 22 3100
```

```
## Done cluster 22
## Cluster size 3100 broken into 1999 1101
## Cluster size 1999 broken into 934 1065
## Done cluster 934
## Done cluster 1065
## Done cluster 1999
## Done cluster 1101
## Done cluster 3100
## Done cluster 3122
## Done cluster 3800
## Cluster size 2048 broken into 1275 773
## Done cluster 1275
## Done cluster 773
## Done cluster 2048
## Done cluster 5848
## Done cluster 581
```

```
## Final global-proteomics matrix: 218 samples × 6429 proteins
```

```
## Running KNN imputation on phosphoproteome (k = 10)...
```

```
## Cluster size 4548 broken into 4081 467
## Cluster size 4081 broken into 1445 2636
## Done cluster 1445
## Cluster size 2636 broken into 342 2294
## Done cluster 342
## Cluster size 2294 broken into 1820 474
## Cluster size 1820 broken into 1212 608
## Done cluster 1212
## Done cluster 608
## Done cluster 1820
## Done cluster 474
## Done cluster 2294
## Done cluster 2636
## Done cluster 4081
## Done cluster 467
```

```
## Final phosphoproteomics matrix: 217 samples × 4548 phosphosites
```

We also build a **samples_matched** helper at this point. Several downstream models (PLS-DA, RF, RSF, the §15 classifier) need the samples table aligned row-for-row with the proteomics matrix, and creating this once upfront eliminates a class of subtle indexing bugs.

```
## samples_matched aligned: 218 rows
```

```
## Class distribution (Y):
```

```
## Y
##         Atypical Teratoid/Rhabdoid Tumor
##                                       12
## Craniopharyngioma, Adamantinomatous Type
##                                       16
##                        Ependymomal Tumor
##                                       32
##                            Ganglioglioma
##                                       18
##                          Medulloblastoma
##                                       22
```

```
##              Pediatric High Grade Gliomas
##                                        25
##               Pediatric Low Grade Gliomas
##                                        93
```

---

4. Exploratory Analysis

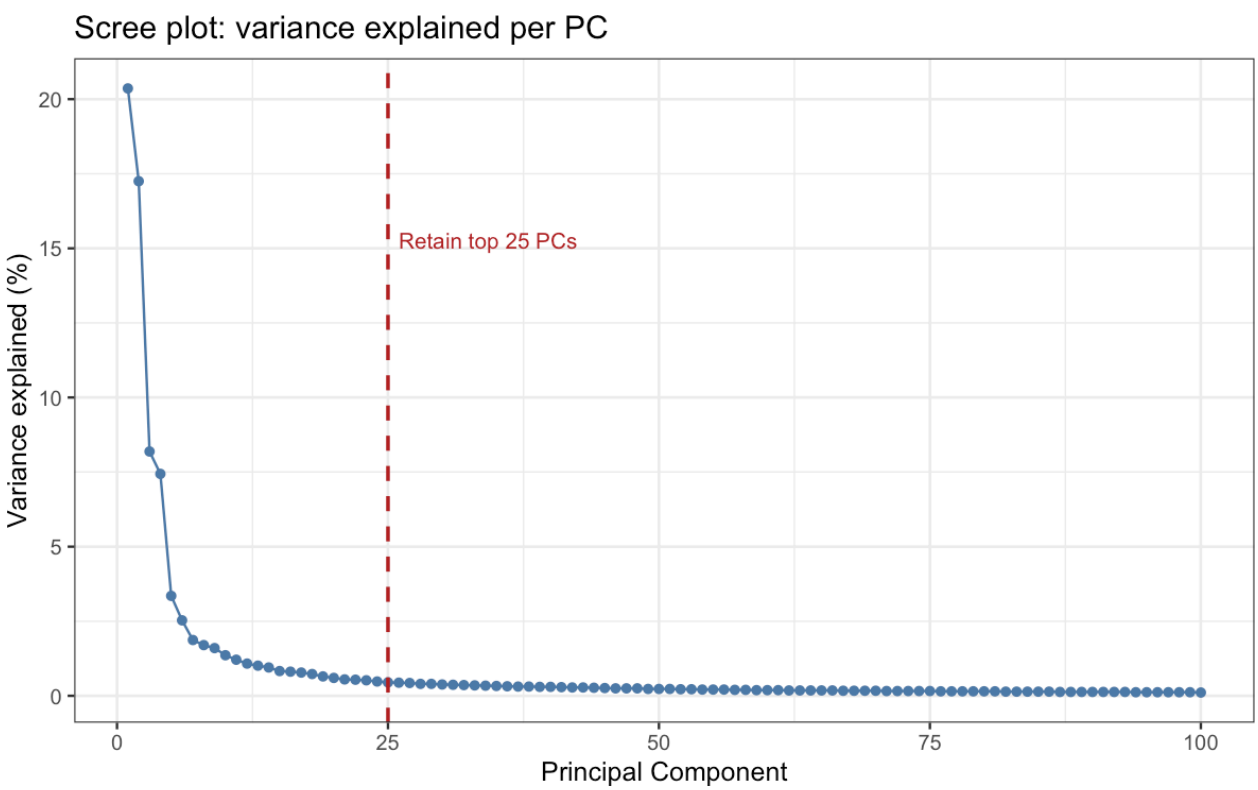
Before committing to NMF, we look at the data with three complementary projections. None of these define our final clusters; their job is only to confirm that meaningful structure exists, and to flag whether that structure is dominated by a small number of subtypes.

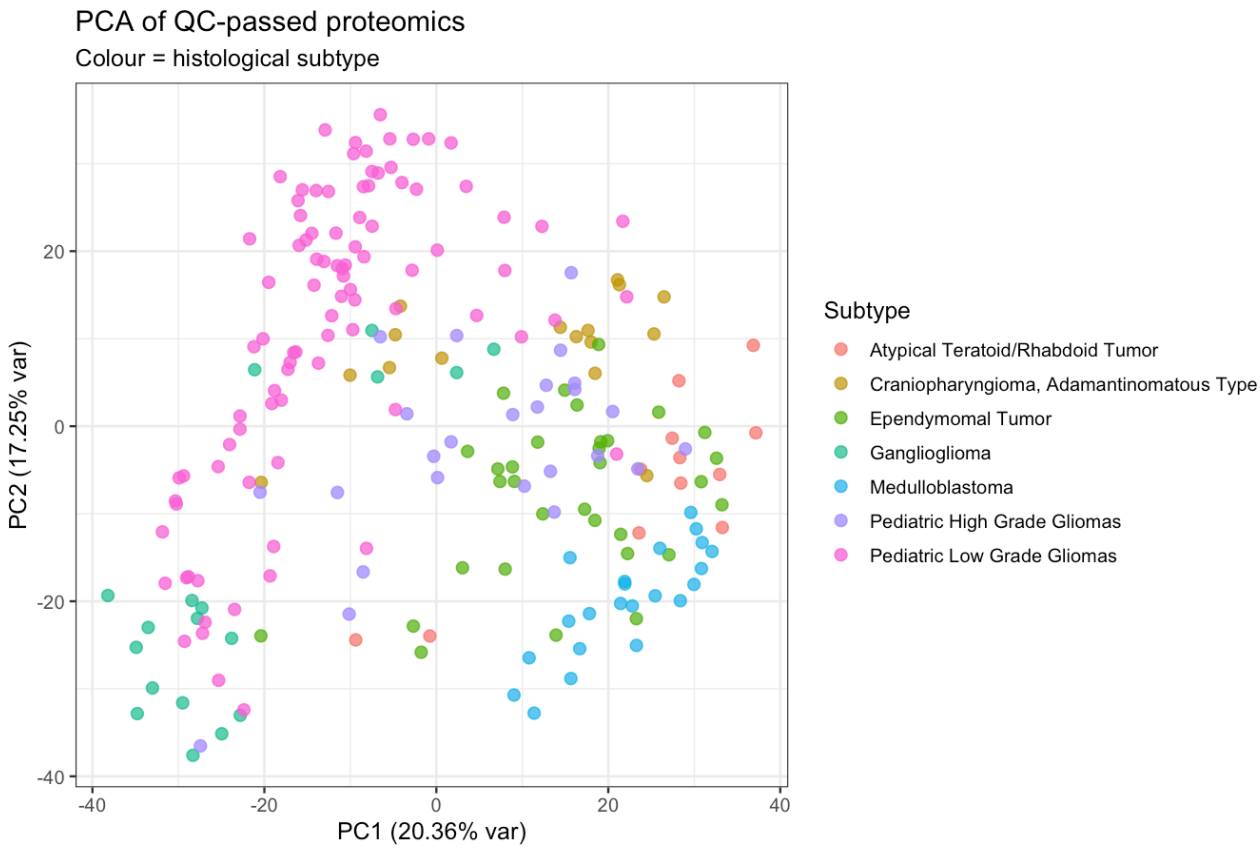
4.1 PCA

PCA gives a first, unsupervised look. We deliberately leave the data un-scaled here because the cBioPortal proteomics file is already z-scored per protein.

```
## Top 3 PCs explain 45.8% of total variance.
```

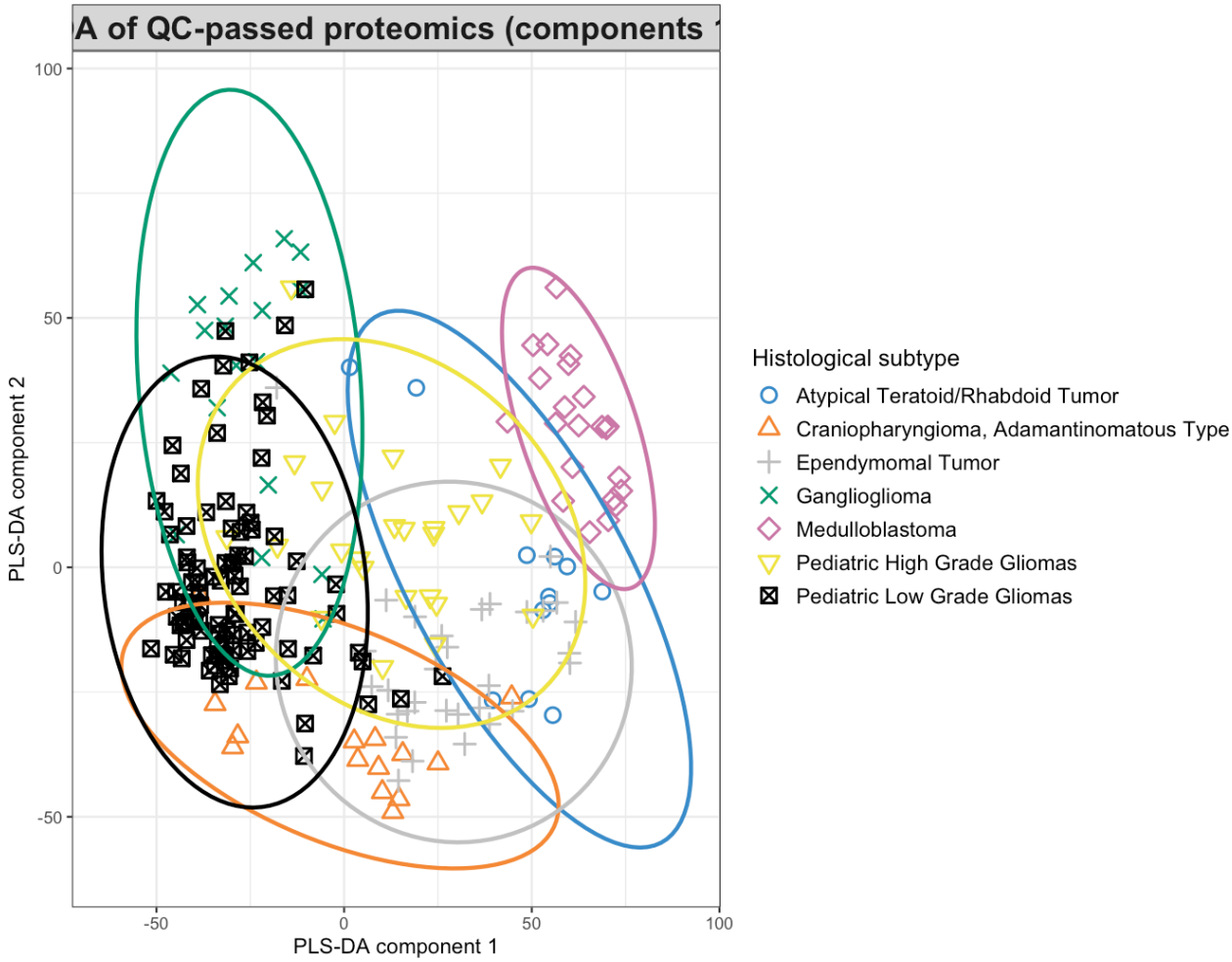
```
## Top 25 PCs explain 76.8% of total variance.
```

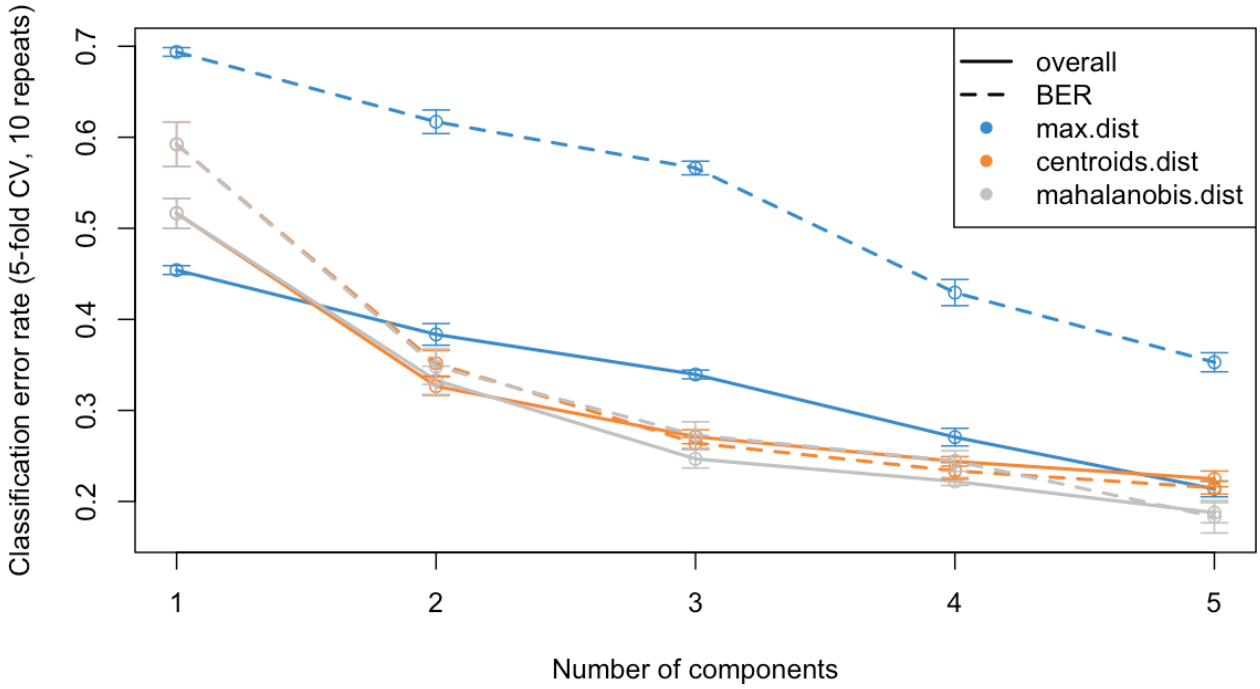


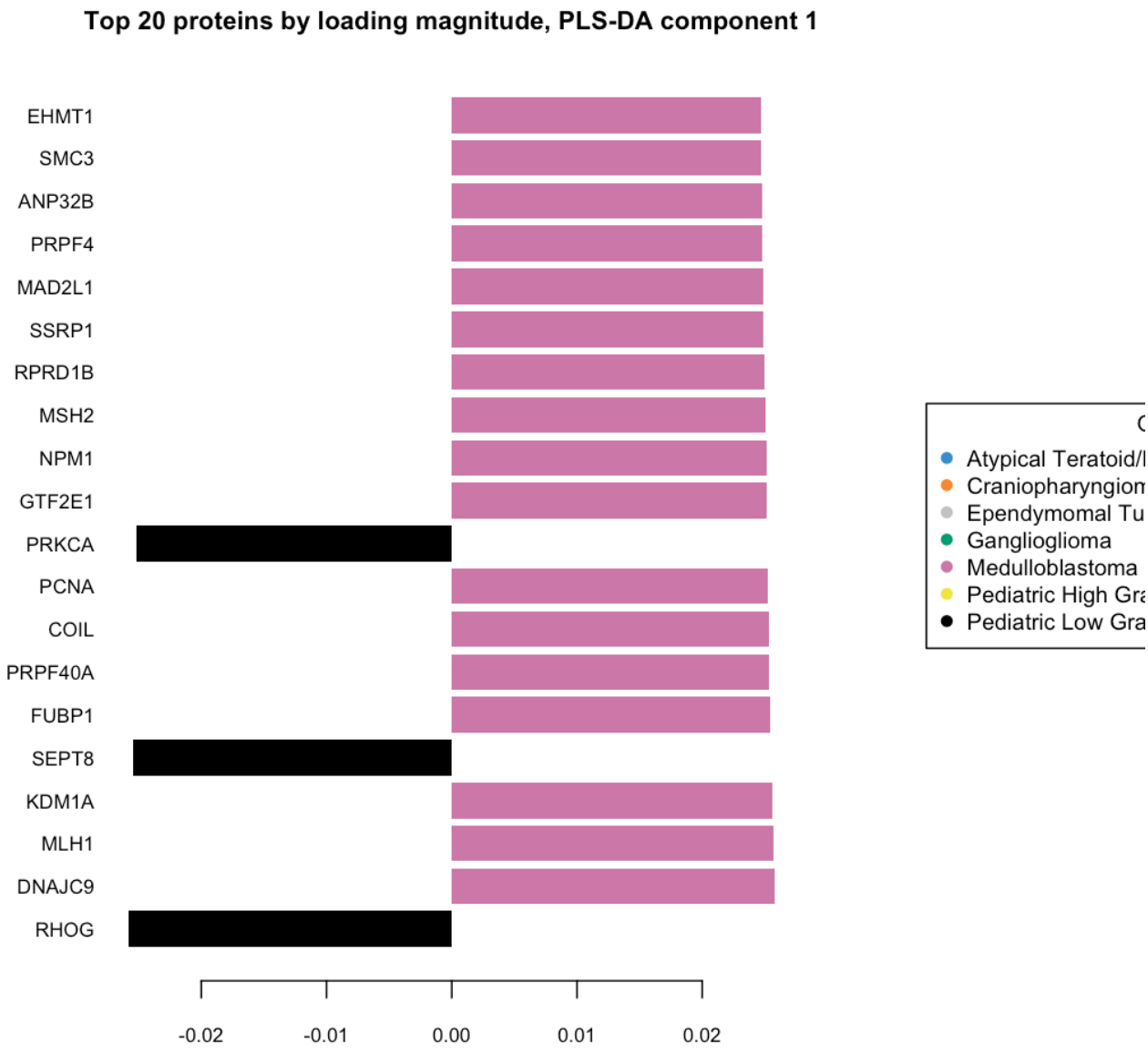


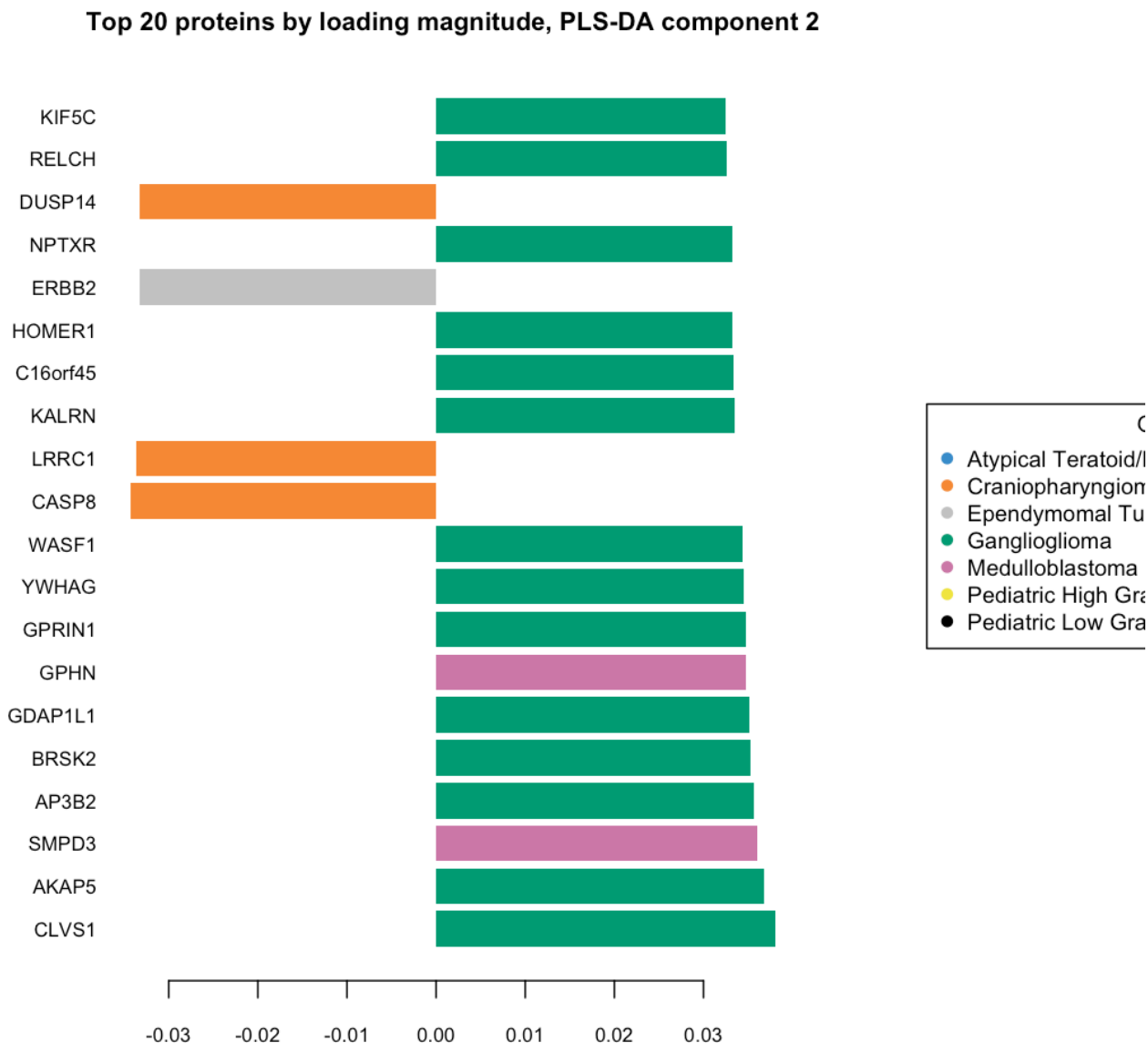
4.2 PLS-DA (supervised projection by histology)

PLS-DA asks the complementary question: if we explicitly use histology as the response, how separable are the subtypes in proteomic space? Strong separation here means histology is a real (but not necessarily complete) source of variation.


A of QC-passed proteomics (components
PLS-DA component 2
PLS-DA component 1
100
50
0
-50
-50
0
50
100
Histological subtype
Atypical Teratoid/Rhabdoid Tumor
Craniopharyngioma, Adamantinomatous Type
Ependymomal Tumor
Ganglioglioma
Medulloblastoma
Pediatric High Grade Gliomas
Pediatric Low Grade Gliomas


Classification error rate (5-fold CV, 10 repeats)
0.2
0.3
0.4
0.5
0.6
0.7
1
2
3
4
5
Number of components
overall
BER
max.dist
centroids.dist
mahalanobis.dist

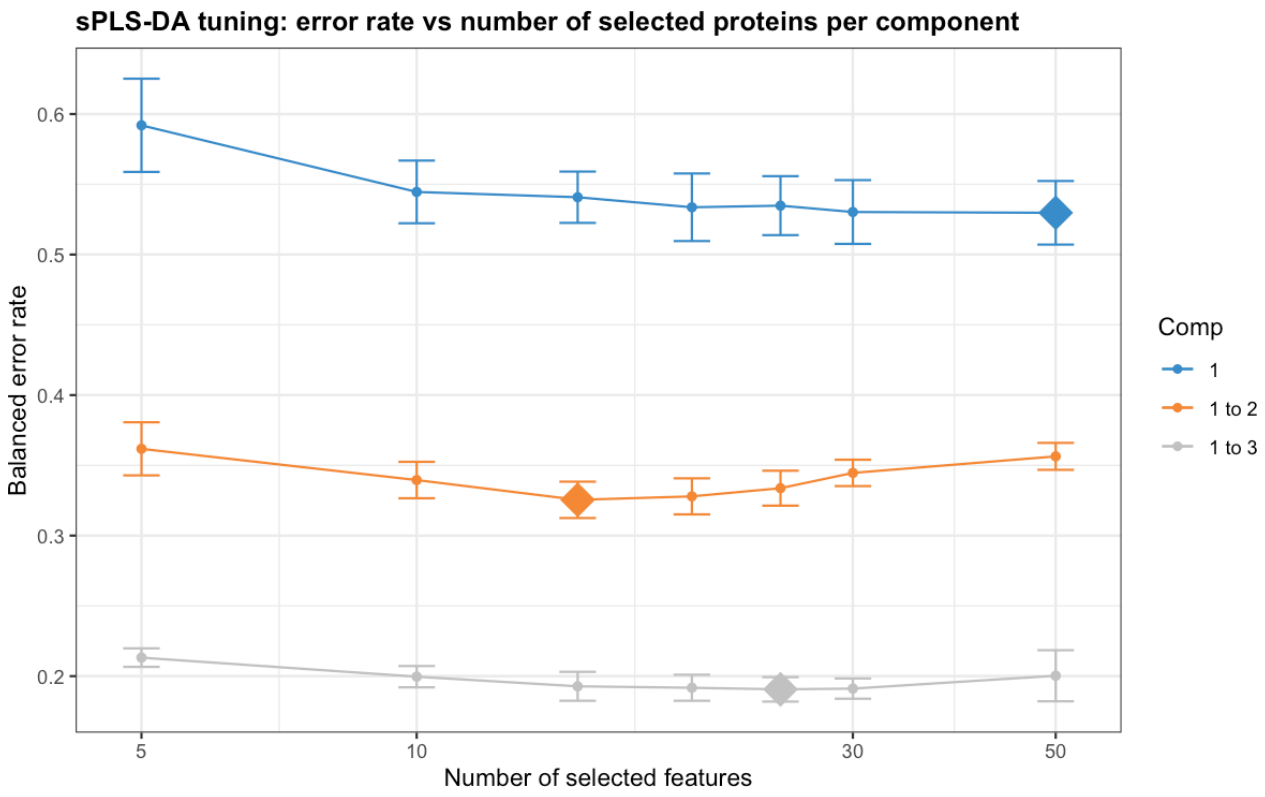

Top 20 proteins by loading magnitude, PLS-DA component 1
EHMT1
SMC3
ANP32B
PRPF4
MAD2L1
SSRP1
RPRD1B
MSH2
NPM1
GTF2E1
PRKCA
PCNA
COIL
PRPF40A
FUBP1
SEPT8
KDM1A
MLH1
DNAJC9
RHOG
-0.02
-0.01
0.00
0.01
0.02
Atypical Teratoid/
Craniopharyngion
Ependymomal Tu
Ganglioglioma
Medulloblastoma
Pediatric High Gra
Pediatric Low Gra

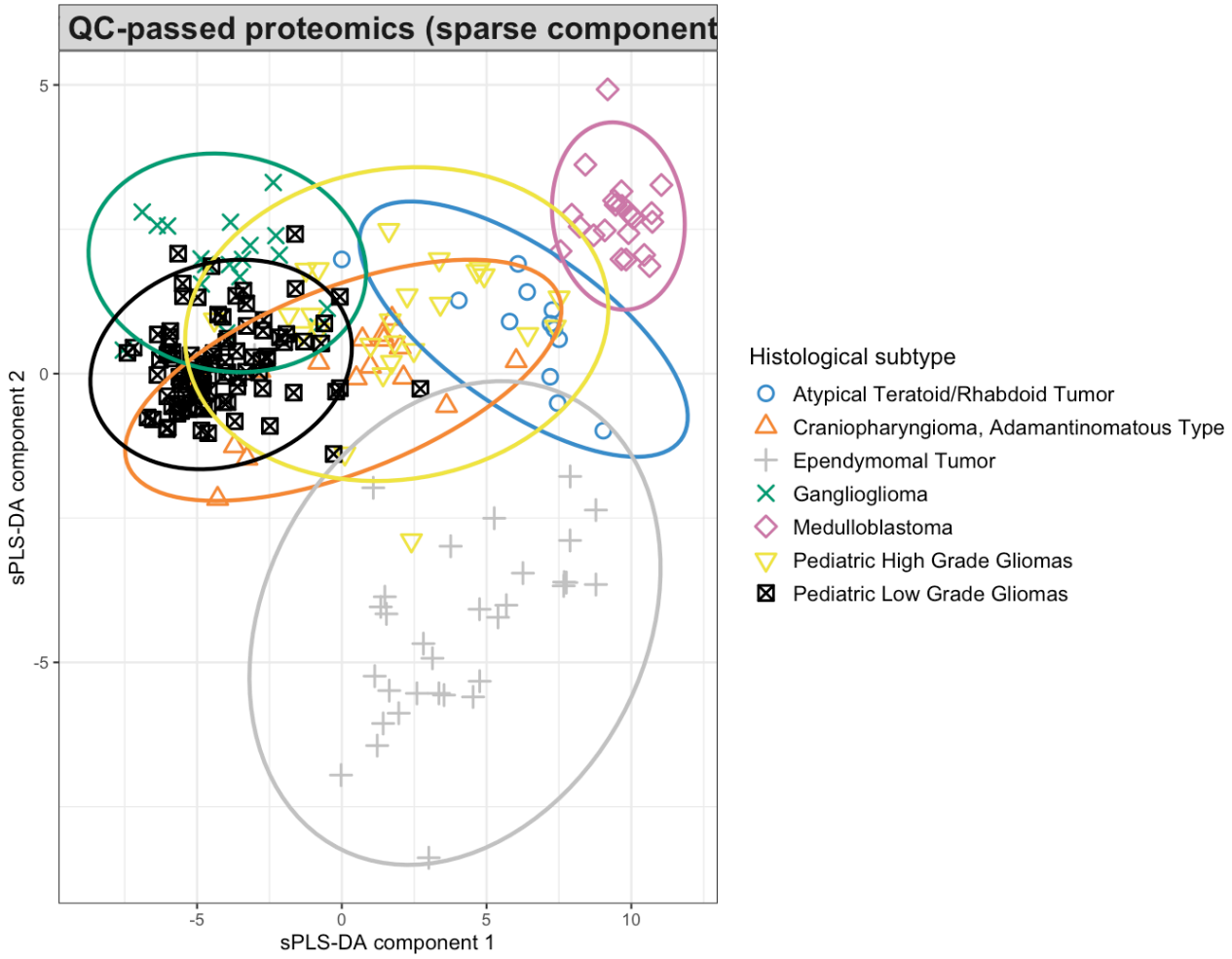


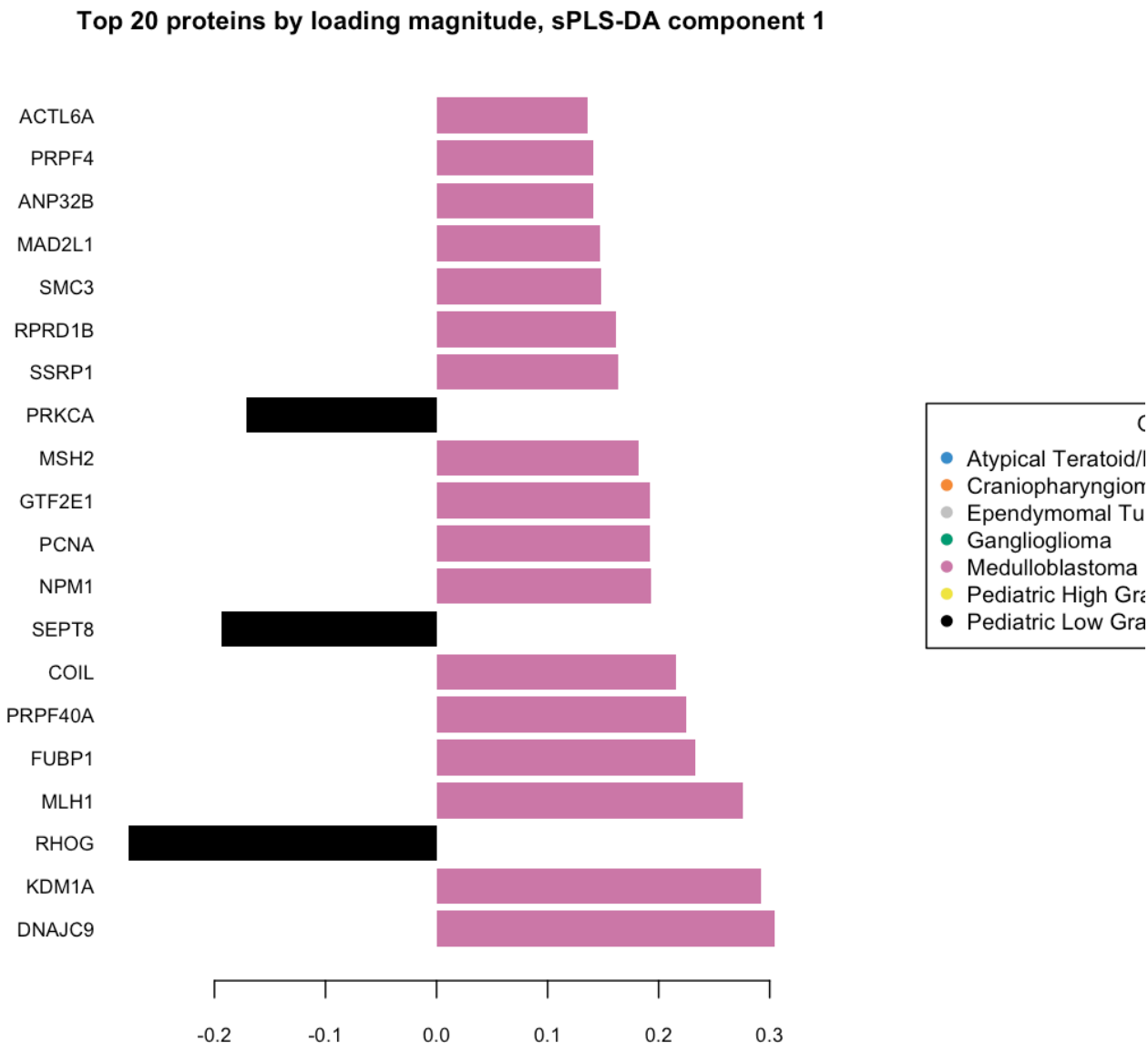
4.3 Sparse PLS-DA (feature selection by histology)

sPLS-DA adds an L1 penalty so that each component is built from only a few proteins. This is a useful first pass at "which proteins distinguish which histology?", and the chosen proteins reappear in the differential expression section later.

```
## comp1 comp2 comp3
##    50    15    25
```

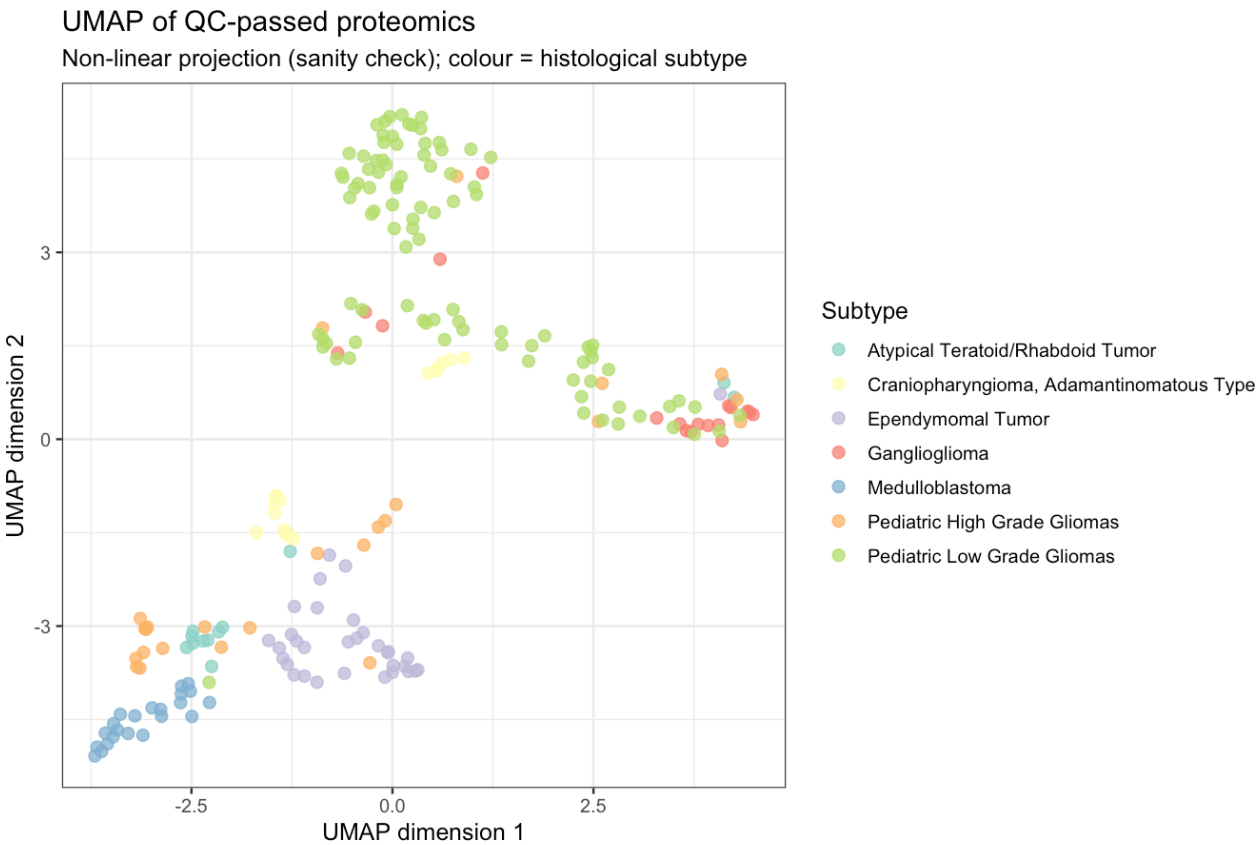

sPLS-DA tuning: error rate vs number of selected proteins per component
Balanced error rate
0.6
0.5
0.4
0.3
0.2
5
10
30
50
Number of selected features
Comp
1
1 to 2
1 to 3


QC-passed proteomics (sparse component
sPLS-DA component 2
sPLS-DA component 1
5
0
-5
-5
0
5
10
Histological subtype
Atypical Teratoid/Rhabdoid Tumor
Craniopharyngioma, Adamantinomatous Type
Ependymomal Tumor
Ganglioglioma
Medulloblastoma
Pediatric High Grade Gliomas
Pediatric Low Grade Gliomas



4.4 UMAP (non-linear sanity check)

UMAP is not used for clustering downstream, it is only a non-linear sanity check that the structure visible in PCA persists when distances are preserved more faithfully.



---

5. Dimensionality Reduction (PCA to NMF input)

Running NMF directly on ~6,000 protein columns is noisy and slow. We compress the matrix to its top `N_PCS` principal components and shift the resulting score matrix to be non-negative (NMF's only hard requirement). The shift is a uniform additive constant and therefore preserves all pairwise distances.

```
## NMF input matrix: 218 samples × 25 PCs
```

```
## Value range: [ 0 , 82.315 ]
```

---

6. Selecting the Number of Clusters (k)

Note on `BEST_K`. Automated metrics on this dataset tend to favour very low k (2–3), which collapses biologically distinct subtypes (e.g. HGG vs LGG) into a single cluster. The default fallback is `BEST_K = 5`.

We sweep k from `K_MIN` to `K_MAX` and evaluate three complementary stability metrics derived from the NMF consensus matrix:

- **Cophenetic correlation**: how faithfully the consensus matrix preserves sample distances (close to 1 = stable).
- **Dispersion**: how bimodal the consensus matrix is (values close to 0 or 1 are good).
- **Silhouette score**: how well-separated clusters are in feature space.

Bug fixed in v3: The NMF R package's convention is features × samples (features as rows, samples as columns). pca_scores_nn is 218 samples × 25 PCs. When you pass it directly (without transpose), NMF treats the 25 PCs as "samples" and the 218 rows as "features" — so it clusters the 25 PCs into k groups, which is meaningless. Passing t(pca_scores_nn) makes it 25 PCs × 218 samples, so NMF correctly treats

the 218 samples as the things being clustered and the 25 PCs as their features. This is what the current code does.

```r
# Slowest cell in the notebook. Set cache=TRUE after first run.
# Note the transpose: NMF treats columns as samples for clustering purposes,
# and the final NMF below uses the same orientation.
nmf_estimates <- nmf(
  t(pca_scores_nn),
  rank = K_MIN:K_MAX,
  nrun = NMF_RUNS,
  seed = 42,
  method = "brunet",
  .options = "v")
```

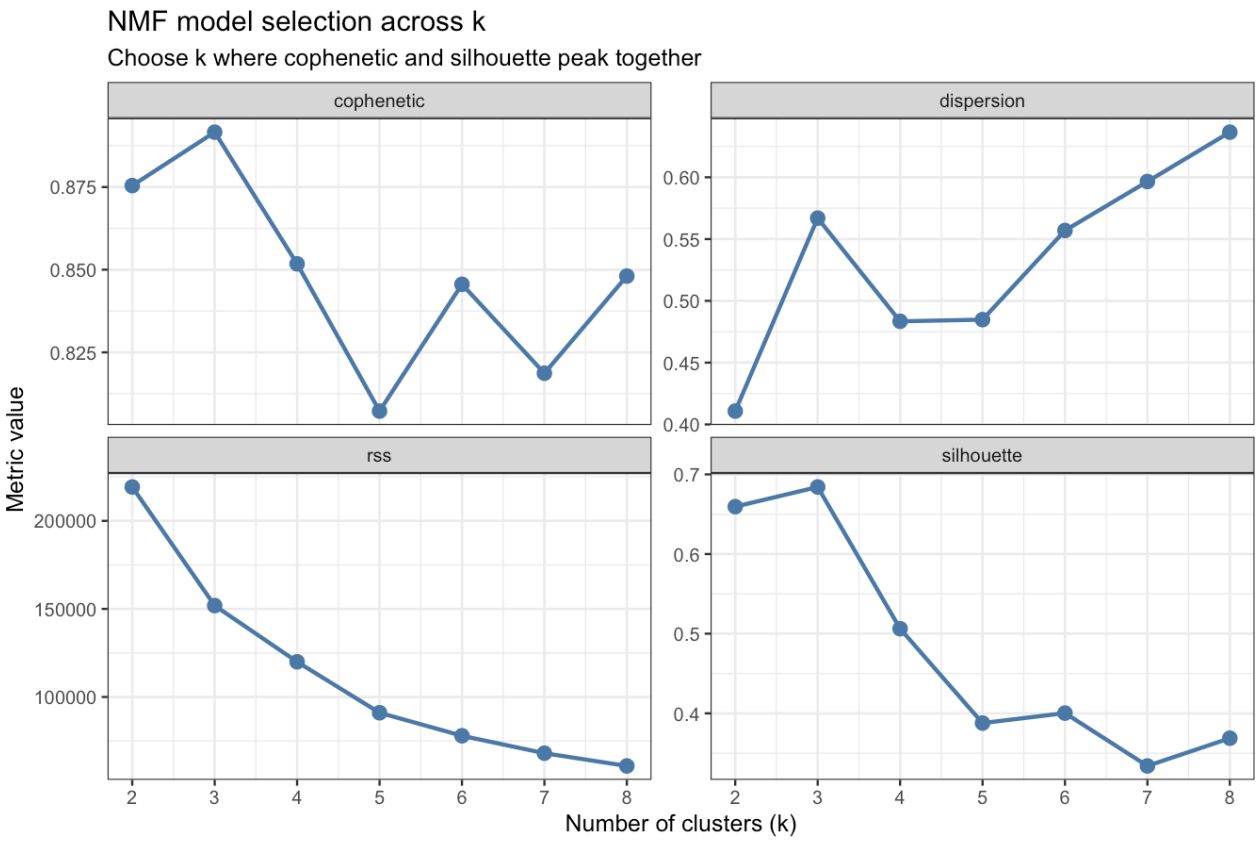
```
## Compute NMF rank= 2  ... + measures ... OK
## Compute NMF rank= 3  ... + measures ... OK
## Compute NMF rank= 4  ... + measures ... OK
## Compute NMF rank= 5  ... + measures ... OK
## Compute NMF rank= 6  ... + measures ... OK
## Compute NMF rank= 7  ... + measures ... OK
## Compute NMF rank= 8  ... + measures ... OK
```

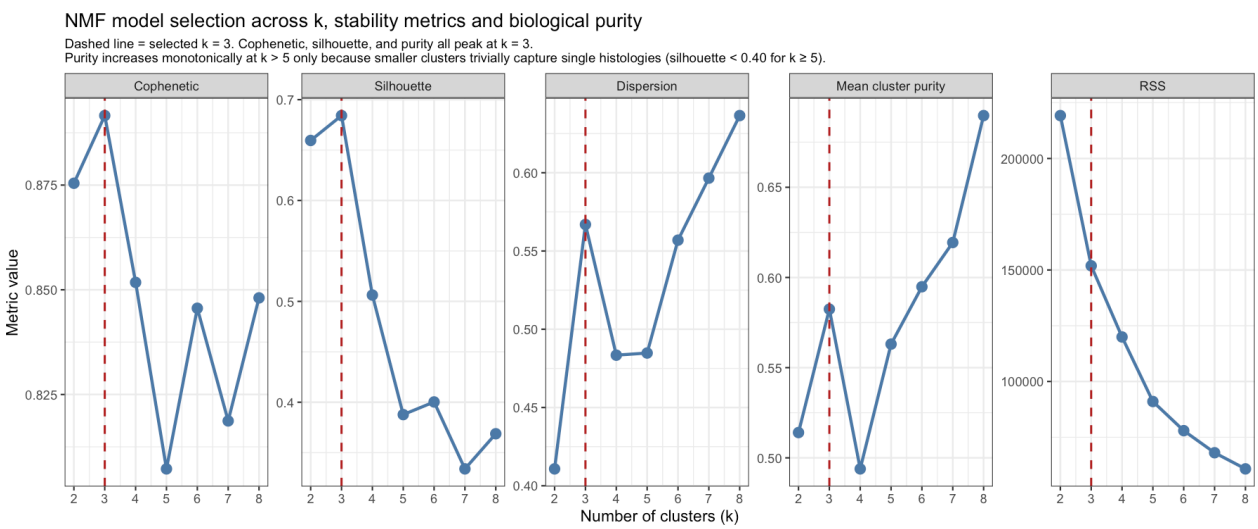
```r
k_metrics <- nmf_estimates$measures |>
  as_tibble() |>
  rename(k = rank, silhouette = silhouette.consensus) |>
  select(k, cophenetic, dispersion, silhouette, rss)

cat("Metrics across k:\n"); print(k_metrics)
```

```
## Metrics across k:
```

```
## # A tibble: 7 x 5
##       k cophenetic dispersion silhouette     rss
##   <dbl>      <dbl>      <dbl>      <dbl>   <dbl>
## 1     2      0.875      0.411      0.660 219221.
## 2     3      0.892      0.567      0.684 151875.
## 3     4      0.852      0.483      0.506 119961.
## 4     5      0.807      0.485      0.388  91029.
## 5     6      0.846      0.557      0.400  77961.
## 6     7      0.819      0.597      0.334  68066.
## 7     8      0.848      0.637      0.369  60830.
```

```r
k_metrics |>
  pivot_longer(-k, names_to = "metric", values_to = "value") |>
  ggplot(aes(x = k, y = value)) +
  geom_line(colour = "#4E79A7", linewidth = 1) +
  geom_point(size = 3, colour = "#4E79A7") +
  facet_wrap(~ metric, scales = "free_y", ncol = 2) +
  scale_x_continuous(breaks = K_MIN:K_MAX) +
  labs(title = "NMF model selection across k",
       subtitle = "Choose k where cophenetic and silhouette peak together",
       x = "Number of clusters (k)", y = "Metric value") +
  theme_bw(base_size = 12)
```






```
## Purity by k:

## # A tibble: 7 x 2
##       k mean_purity
##   <int>       <dbl>
## 1     2       0.514
## 2     3       0.582
## 3     4       0.494
## 4     5       0.563
## 5     6       0.595
## 6     7       0.619
```

```
## 7     8       0.690

## Data-driven k = 3  (silhouette = 0.684, cophenetic = 0.892)
```

---

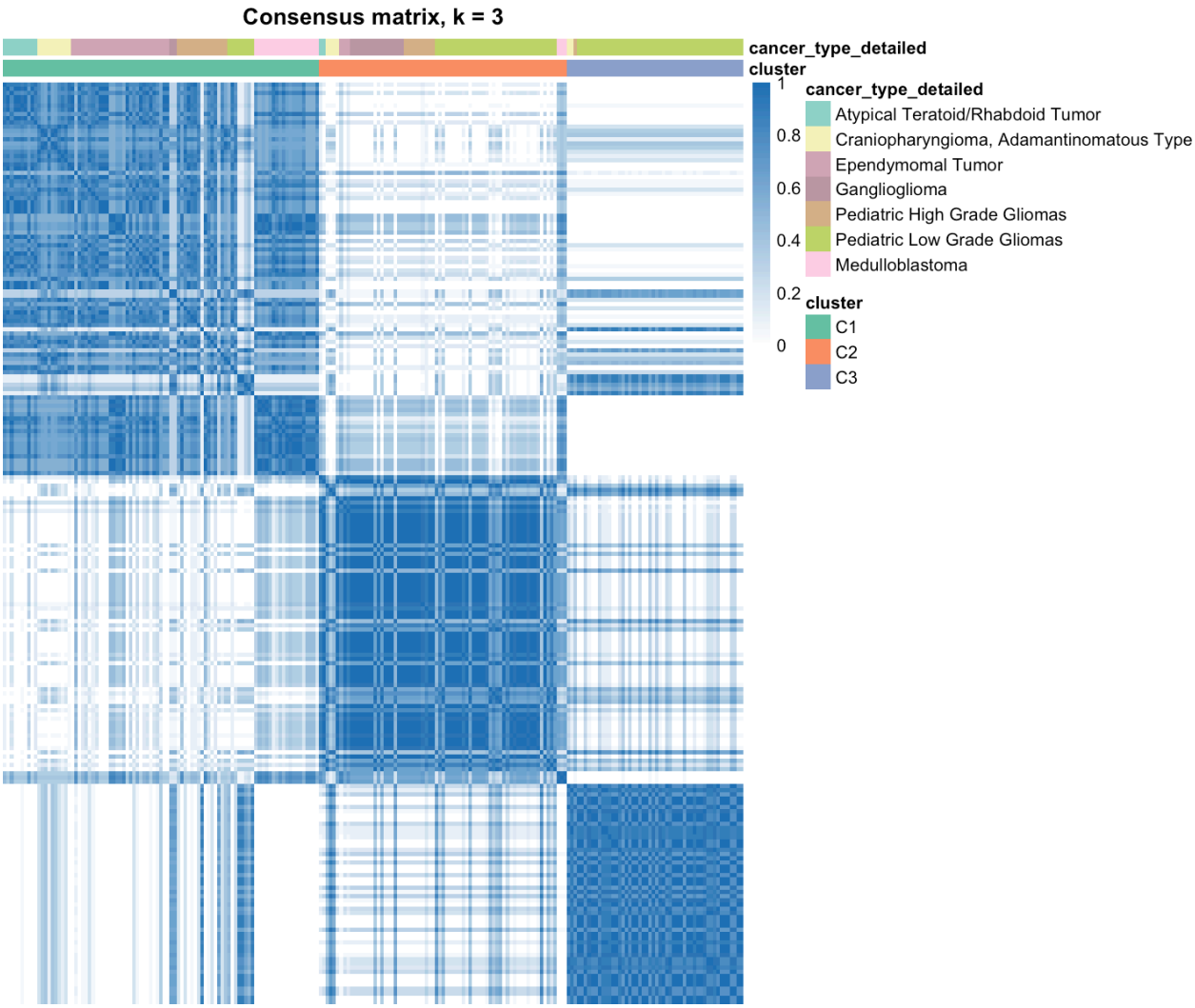
7. Consensus NMF Clustering

With `BEST_K` chosen, we run a final NMF and extract the consensus matrix and cluster labels.

Interpreting the consensus heatmap:

- Dark blue diagonal blocks indicate patients always cluster together (stable).
- Light off-diagonal regions indicate patients from different clusters never co-cluster (good separation).
- Mixed histologies within a cluster indicate cross-boundary molecular grouping, a key finding of the original paper.

```
## Runs: |                                                          Runs: |
## System time:
##    user  system elapsed
##  12.232  12.778  28.632

## cons_matrix dim: 218 218

## Cluster distribution:

## cluster_labels
##  1  2  3
## 93 73 52
```



---

8. Cluster Annotation

Before running biology, we assemble a single tidy table that joins cluster labels with the patient-level survival and demographic fields. Doing this once here means later chunks can stay focused on modelling.

```
## Cluster sizes:
```

```
## # A tibble: 3 x 2
##   cluster     n
##     <int> <int>
## 1       1    93
## 2       2    73
## 3       3    52
```

```
## Missing os_months: 7
```

```
## Missing dfs_months: 7
```

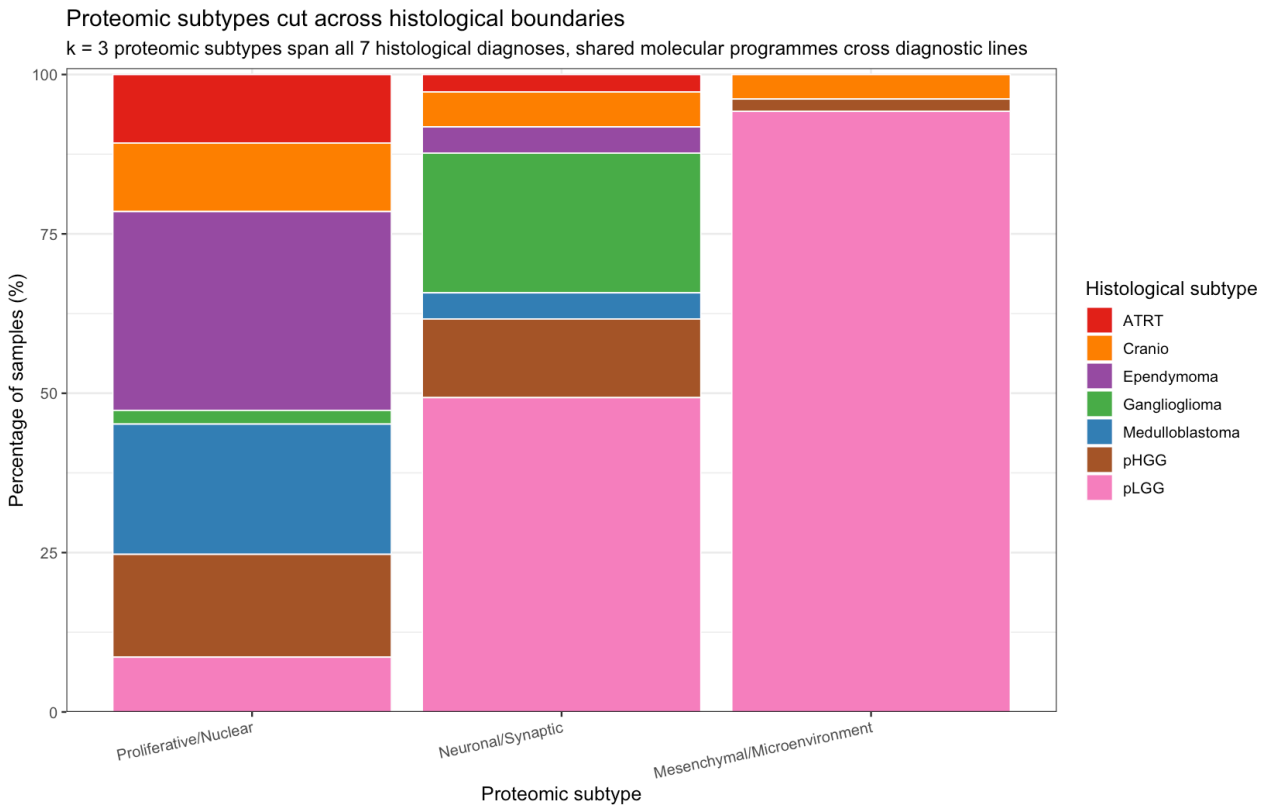
```
## # A tibble: 3 x 6
##   cluster     n n_events event_rate median_os mean_os
##     <int> <int>    <int>      <dbl>     <dbl>   <dbl>
```

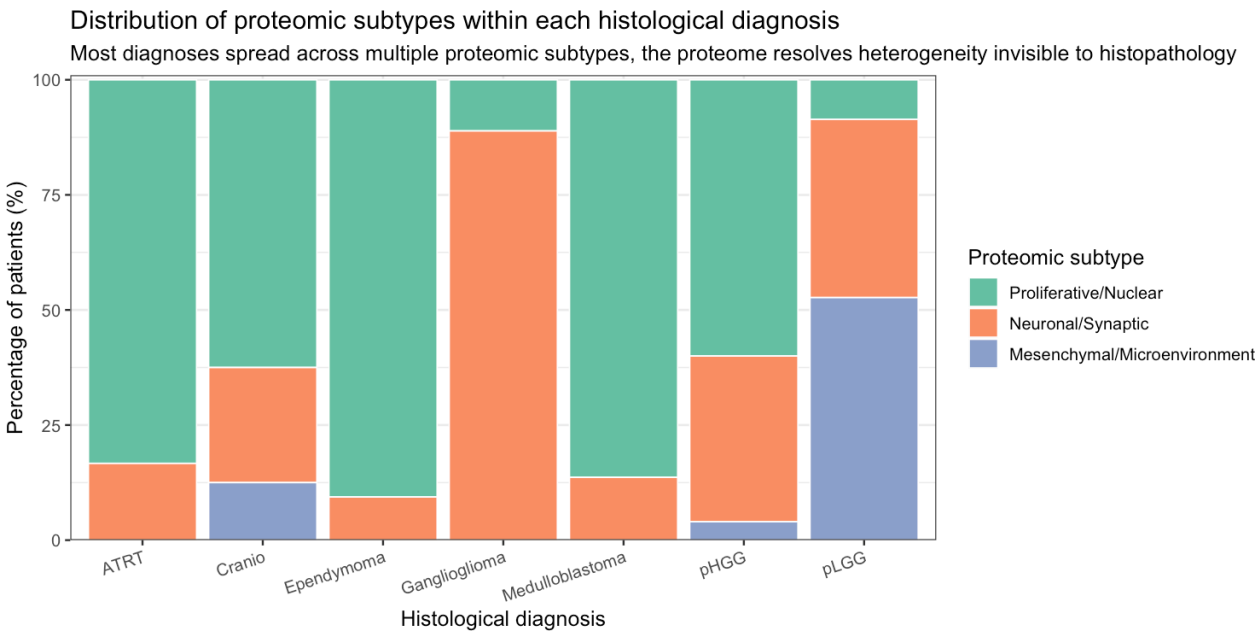
```
## 1        1     92        33        35.9       32.5     41
## 2        2     73        13        17.8       34       47.4
## 3        3     46         1         2.2       38.5     52.9
```

---

9. Histological Composition per Cluster

Do the proteomics-derived clusters recapitulate known histology, or do they reveal cross-boundary groupings? This needs to be answered before survival analysis: a cluster that is 100% Medulloblastoma is just re-discovering the diagnosis, while a cluster that mixes histologies is genuinely informative.






```
##
##      Atypical Teratoid/Rhabdoid Tumor Craniopharyngioma, Adamantinomatous Type
##   C1                              10                                       10
##   C2                               2                                        4
##   C3                               0                                        2
##
##      Ependymomal Tumor Ganglioglioma Medulloblastoma
##   C1                29             2              19
##   C2                 3            16               3
##   C3                 0             0               0
##
##      Pediatric High Grade Gliomas Pediatric Low Grade Gliomas
##   C1                           15                          8
##   C2                            9                         36
##   C3                            1                         49
```

```
##
##      Atypical Teratoid/Rhabdoid Tumor Craniopharyngioma, Adamantinomatous Type
##   C1                             0.11                                     0.11
##   C2                             0.03                                     0.05
##   C3                             0.00                                     0.04
##
##      Ependymomal Tumor Ganglioglioma Medulloblastoma
##   C1              0.31          0.02            0.20
##   C2              0.04          0.22            0.04
##   C3              0.00          0.00            0.00
##
##      Pediatric High Grade Gliomas Pediatric Low Grade Gliomas
##   C1                         0.16                        0.09
##   C2                         0.12                        0.49
##   C3                         0.02                        0.94
```

```
## # A tibble: 9 x 3
## # Groups:   cluster [3]
##   cluster subtype                                       n
```

```
##   <fct>   <fct>                                   <int>
## 1 C1      Ependymomal Tumor                          29
## 2 C1      Medulloblastoma                            19
## 3 C1      Pediatric High Grade Gliomas               15
## 4 C2      Pediatric Low Grade Gliomas                36
## 5 C2      Ganglioglioma                              16
## 6 C2      Pediatric High Grade Gliomas                9
## 7 C3      Pediatric Low Grade Gliomas                49
## 8 C3      Craniopharyngioma, Adamantinomatous Type    2
## 9 C3      Pediatric High Grade Gliomas                1
```

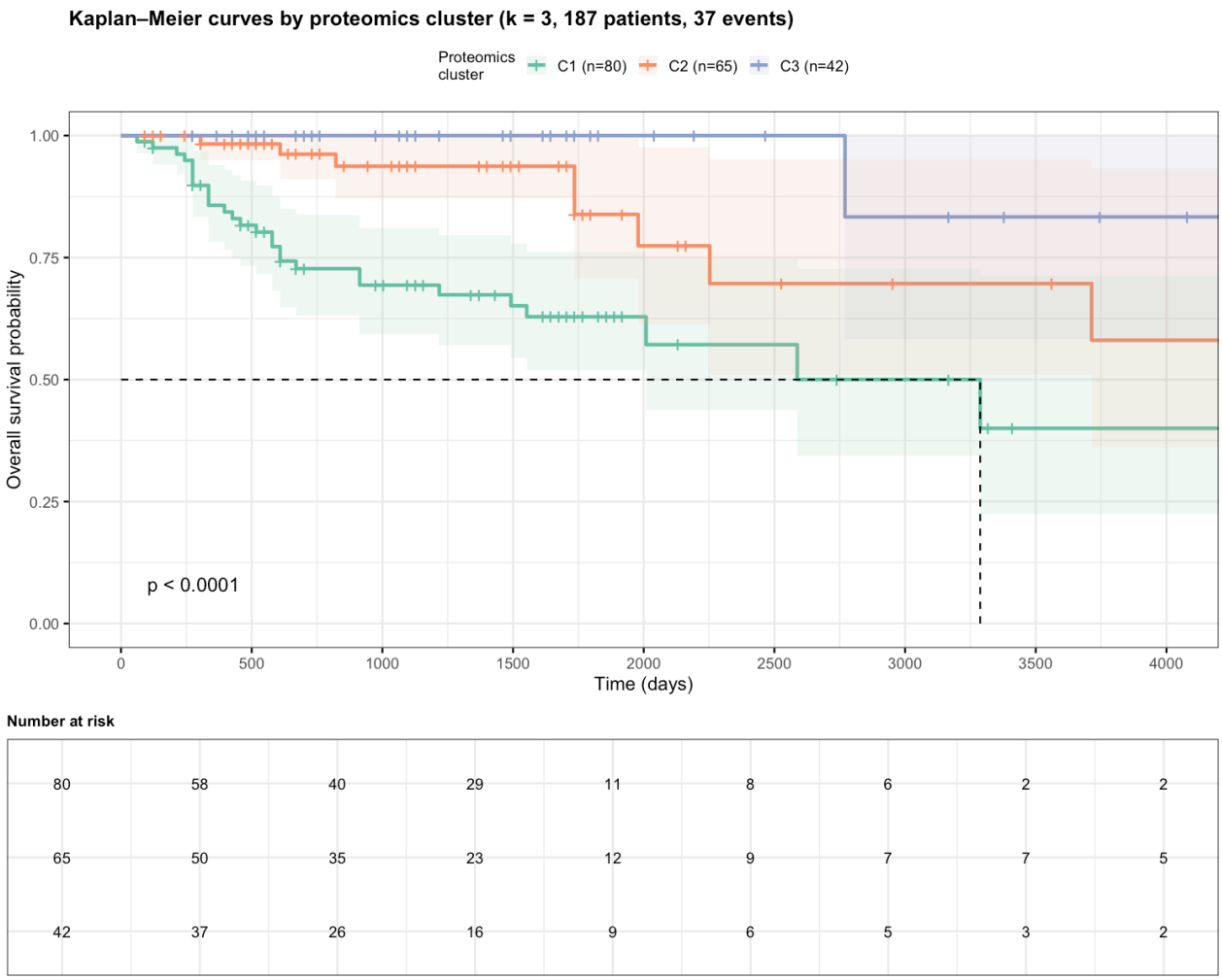
What to look for: Clusters dominated by a single histological subtype mostly recapitulate known diagnoses. Clusters that mix several subtypes, for example LGG alongside Ependymoma, are more interesting (and common here): they suggest a shared underlying molecular programme that cuts across classical histological boundaries, which is the central finding of Petralia et al. (2020).

---

10. Survival Analysis (Kaplan–Meier)

A key clinical test: do the identified subgroups have different survival outcomes? We use Kaplan–Meier curves and the log-rank test. A significant p-value (< 0.05) means the clusters capture clinically meaningful biology, not just abstract molecular similarity.

```
## Patients: 187 Events: 37
```




```
##            records events median 0.95LCL 0.95UCL
## cluster=C1      80     28    108      66      NA
## cluster=C2      65      8     NA     122      NA
## cluster=C3      42      1     NA      NA      NA
```

---

11. Cox Proportional Hazards Model

The KM curves answer "are the clusters different?". The Cox model answers the next question: is that difference still there after we adjust for what we already knew about the patient? If `cluster` retains a significant hazard ratio after adjustment, the proteomic structure is contributing risk information beyond clinical baseline.

In this version the clinical baseline is no longer just age + sex. We add WHO grade, extent of resection, treatment received (radiation, chemotherapy), and the canonical pediatric brain tumour driver flags (`braf_status`, `h3f3a_ctnnb1_status`, etc.). Beating this baseline is a much more meaningful demonstration that proteomics adds prognostic information than beating age + sex alone.

```
## Cox model: 187 patients (37 events), 3 clusters
```

```
##
## Cox model coefficients (extended baseline)
```

```
##                                        term log_HR     HR    SE z_score
```

```
##                                   clusterC2  0.006  1.006 0.489   0.013
##                                   clusterC3 -1.328  0.265 1.142  -1.163
##                                         age  0.000  1.000 0.000   0.750
##                                     sexMale -0.218  0.804 0.370  -0.589
##                           grade_binHighGrade  2.641 14.028 0.700   3.772
##  extent_resectionGross/Near total resection -1.348  0.260 0.975  -1.382
##                         extent_resectionN/A     NA     NA 0.000      NA
##              extent_resectionNot Applicable -1.213  0.297 1.093  -1.110
##           extent_resectionPartial resection -0.715  0.489 0.930  -0.769
##                              radiation_ynYes -0.624  0.536 0.524  -1.192
##                           chemotherapy_ynYes -0.131  0.877 0.410  -0.320
##                   any_known_driverAnyDriver -0.636  0.530 0.804  -0.791
##   p_value sig
##      0.99
##     0.245
##     0.453
##     0.556
##  0.000162 ***
##     0.167
##        NA
##     0.267
##     0.442
##     0.233
##     0.749
##     0.429
```

```
## # A tibble: 11 x 2
##    Metric                      Value
##    <chr>                       <chr>
##  1 Concordance (Harrell's C)  0.87
##  2 Concordance SE             0.024
##  3 Likelihood ratio test ( ²) 62.996
##  4 LRT p-value                2.56e-09
##  5 Wald test ( ²)             41.25
##  6 Wald p-value               2.18e-05
##  7 Score (log-rank) test ( ²) 76.319
##  8 Score p-value              7.56e-12
##  9 AIC                        283.2
## 10 n (complete cases)         184
## 11 Events                     36
```

Interpreting the Cox output:

- HR > 1 = higher hazard (worse survival) relative to the reference level.
- HR < 1 = lower hazard (better survival).
- A small cluster with very few events (the Hauck–Donner / complete-separation case) will produce HR estimates with extreme SEs that should be ignored; the concordance statistic is the trustworthy global summary.
- A cluster effect surviving adjustment for the extended clinical baseline is the descriptive evidence that the proteomic stratification is prognostically informative beyond standard-of-care variables. We deliberately stop at description and do not build a per-patient predictive risk score on this cohort, with only 36 OS events the events-per-parameter ratio is too low to support an honest predictive claim. The §14 therapeutic-vulnerability arm and the §15 molecular classifier sidestep this small-event problem by addressing different questions.

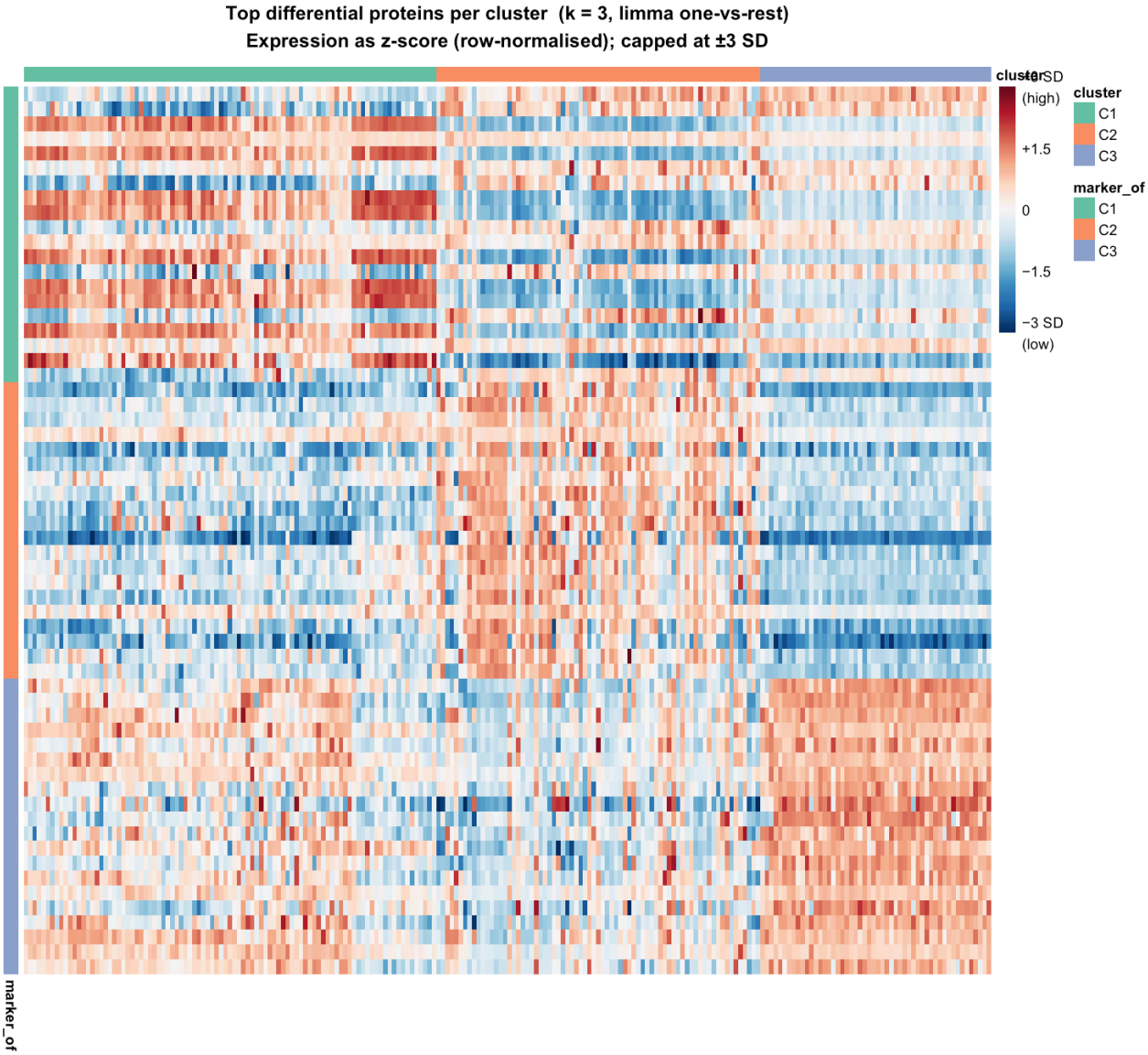
---

12. Differential Protein Expression (limma one-vs-rest)

We identify proteins significantly up-regulated in each cluster relative to all other clusters combined. These become the molecular fingerprints of each subtype, and they also seed candidate cluster-marker shortlists used by the §14 therapeutic-target panel.

`limma` is well-validated for mass-spectrometry proteomics data (continuous, approximately normal after z-scoring) and handles the moderate sample sizes of this dataset well.

```
## [1] 60
```

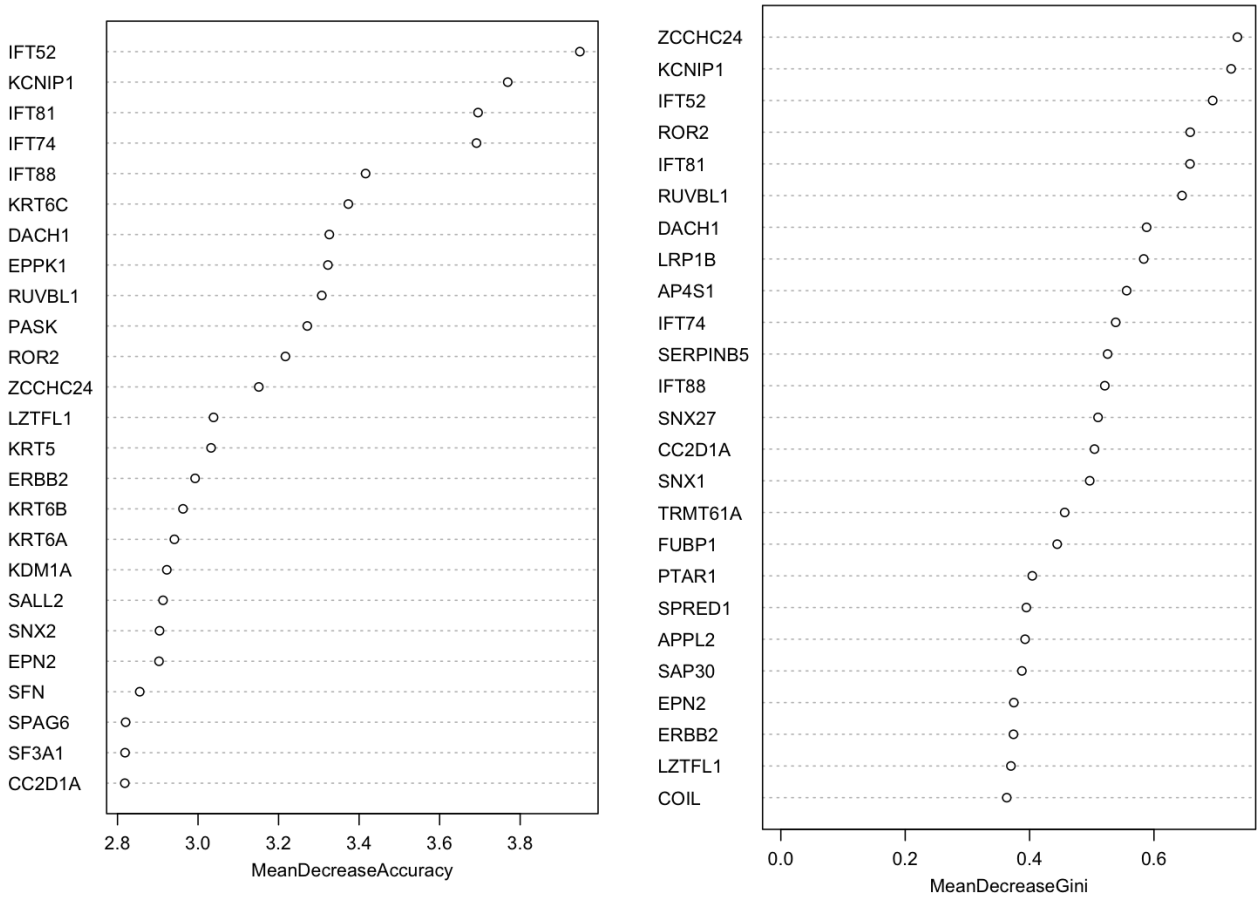



```
## Top 10 upregulated marker proteins per cluster:
##
##   Cluster 1: SEPT8, NCAM2, SON, ARHGAP39, RBM25, RCAN1, HSPA12A, ACIN1, PRPF40A, TMOD2
##
##   Cluster 2: IGSF8, LYNX1, PPP3CA, MDH1, TUBB4A, GNAO1, ADAM22, CNTNAP1, PHYHIP, NFASC
##
##   Cluster 3: HIP1, LRP1, PHLDB1, SLC12A9, CC2D1A, SNX1, SLC9A7, NLGN1, CSPG4, SPRED1
```

13. Random Forest (classification by histology)

These two ensemble models give a model-free, descriptive ranking of which individual proteins carry the strongest signal, first against histological subtype (Random Forest classifier) and then against OS event timing (Random Survival Forest). They do not produce a per-patient risk score; their role is to surface candidate proteins that subsequently feed into the cluster-marker tables in §14.

Top 25 proteins by Random Forest importance (histology classification)



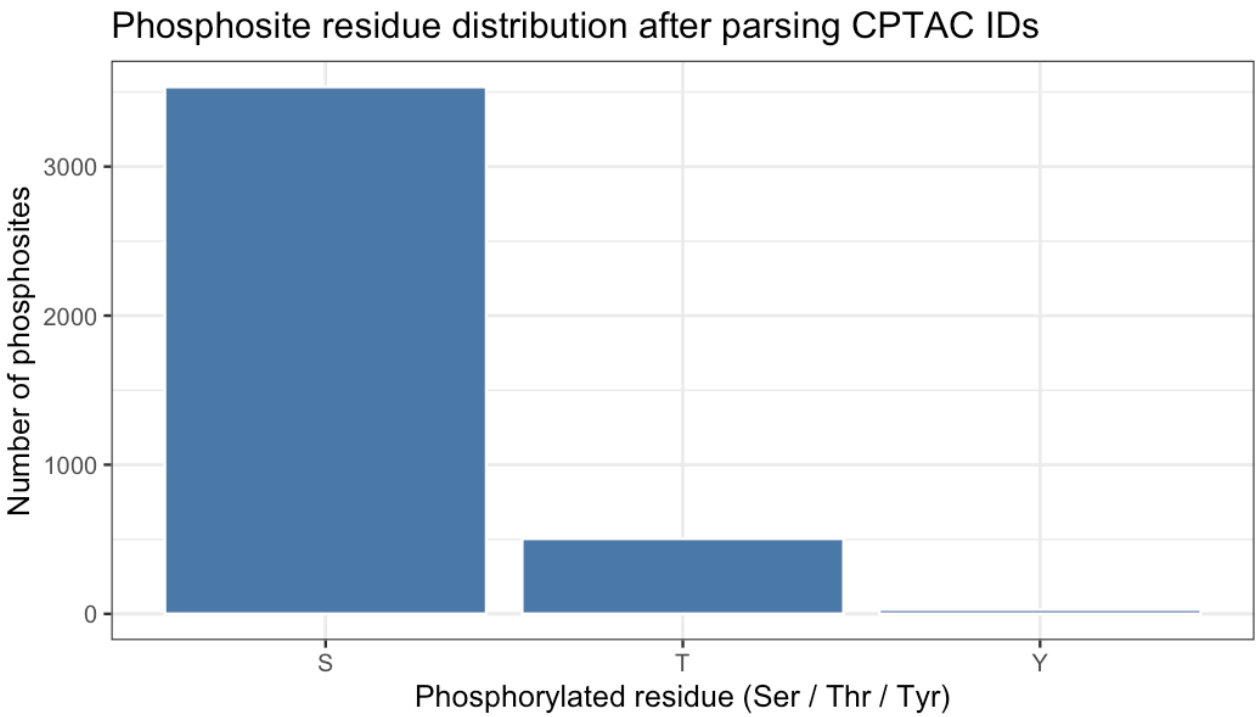
14. Therapeutic Target Discovery

This section is the precision-oncology deliverable. For each NMF cluster identified in §7, we build a per-cluster panel of candidate therapeutic targets supported by multiple, independent lines of evidence: cluster-specific overexpression (limma; pulled from §12), inferred kinase activity (KSEA on per-cluster phospho contrasts, substrates from OmniPath), brain cell-line dependency (DepMap CRISPR essentiality), and druggability (OpenTargets clinical-trial / approved-drug lookup). A target that scores well on at least two of these is a defensible candidate for a subtype-specific therapy panel.

14.1 Per-cluster phosphosite contrasts (limma)

Phospho rows in CPTAC follow `<GENE>_<start>_<end>_<x>_<y>_<RES><pos>`, e.g. `ALAD_214_215_1_1_S215`. We parse those into (gene, residue type, position) so we can join phosphosites to OmniPath kinase–substrate entries.

```
## Phospho ID parser: 89.4% of 4548 sites parsed cleanly
```

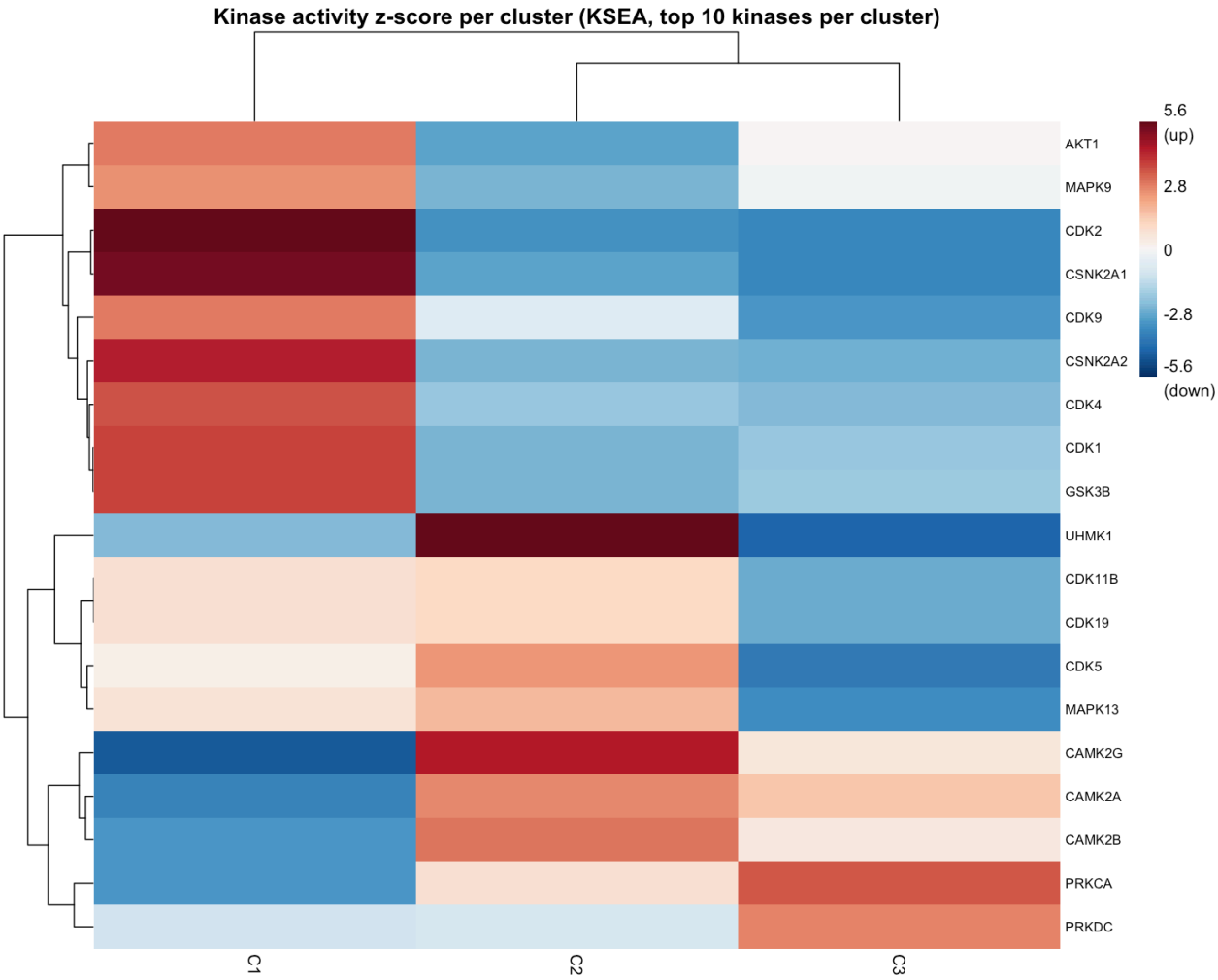



```
## Phospho one-vs-rest contrasts on 217 samples × 4548 phosphosites

## Phospho DE table: 13644 rows

## # A tibble: 5 x 8
##   cluster phosphosite         gene_symbol residue_type residue_offset logFC_phos
##   <chr>   <chr>               <chr>       <chr>                 <int>      <dbl>
## 1 C1      ALAD_214_215_1_1_S~ ALAD        S                       215    -0.538
## 2 C1      ALDOA_36_39_1_1_S36 ALDOA       S                        36    -1.04
## 3 C1      ALDOA_36_39_1_1_S39 ALDOA       S                        39    -0.866
## 4 C1      ALDOA_46_52_1_1_S46 ALDOA       S                        46    -0.0574
## 5 C1      ANK1_1684_1693_1_1~ ANK1        S                      1686    -0.104
## # i 2 more variables: P.Value <dbl>, q <dbl>
```

14.2 Kinase activity per cluster (KSEA)

KSEA assigns each kinase a z-score per cluster from the mean log fold-change of its known phospho substrates relative to background. Substrates come from OmniPath, which aggregates PhosphoSitePlus, Signor, and others. Kinases with fewer than `KSEA_MIN_SUBSTRATES` matched substrates are dropped to avoid noisy estimates.

```
## OmniPath kinase-substrate entries (S/T/Y phosphorylation): 39391

## Phosphosites with at least one OmniPath kinase: 659 / 4548 (14.5%)

## KSEA results: 258 kinase × cluster rows
```



14.3 Cluster-marker shortlist (overexpressed proteins)

Re-run the §12 limma machinery to recover the full statistics table per cluster (not just the top-20 names), then filter to genes that are up-regulated only, these are candidates for inhibition.

```
## Cluster marker shortlist: 150 rows across 3 clusters
```

```
## # A tibble: 3 x 2
##   cluster n_candidates
##   <chr>          <int>
## 1 C1                50
## 2 C2                50
## 3 C3                50
```

14.4 DepMap brain-line dependency

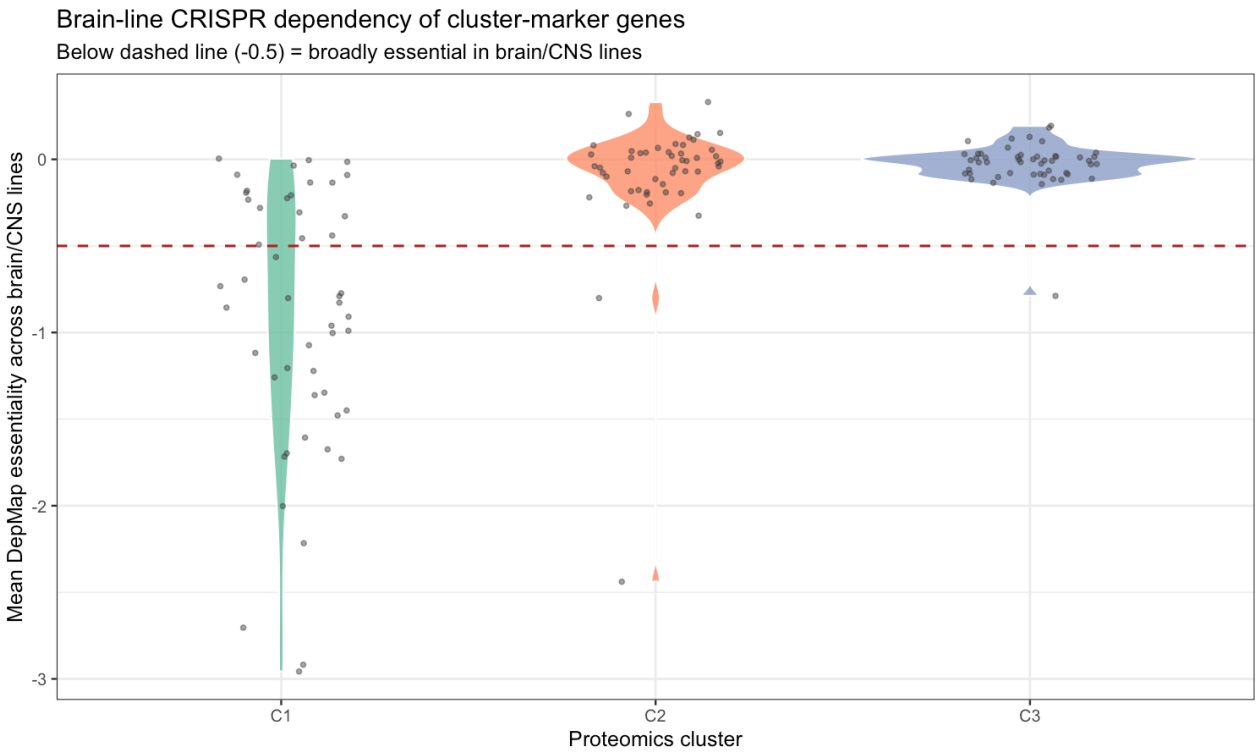
Cross-reference the cluster-marker shortlist with CRISPR essentiality scores from DepMap, restricted to brain/CNS cell lines. A gene that is both upregulated in a cluster AND broadly essential in brain lines is a strong candidate for therapeutic inhibition.

```
## Loading DepMap files...
```

```
## Brain/CNS DepMap cell lines: 127
```

```
## dep_summary: 18435 genes across 127 brain lines
```

```
## Gene overlap: 149 / 150 marker genes found in DepMap
## Marker genes with DepMap entry: 149 / 150 (99.3%)
## Cluster-marker x DepMap-essential overlaps: 34
```



14.5 Druggability annotation (OpenTargets)

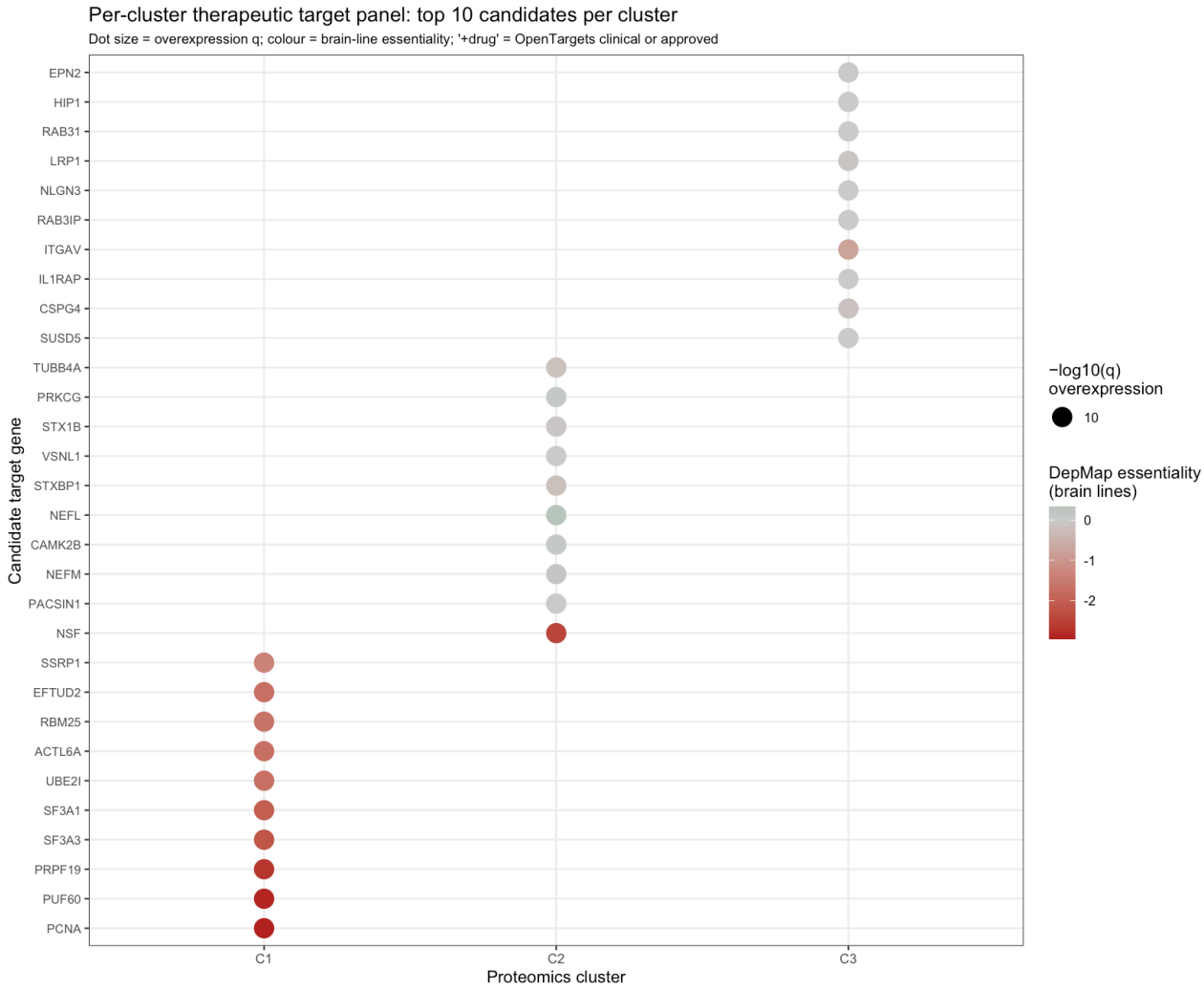
Cross-reference candidate genes with OpenTargets to flag which already have approved or clinical-trial drugs.

```
## Candidate genes for druggability lookup: 30
## Symbols mapped to Ensembl IDs: 32 / 30
## Druggable hits (raw CHEMBL IDs): 1
## Resolving 1 unique CHEMBL IDs to drug names...
## Names resolved: 1 / 1
## Druggable hits returned: 1
##    gene   drug_name    chembl_id max_phase
## 1 PRKD1 MIDOSTAURIN CHEMBL608533         4
```

14.6 Integrated per-cluster therapeutic panel

Combine the four evidence streams into a single ranked panel per cluster. The composite score is a sum of standardised components (each missing stream contributes 0), keeping the ranking robust when individual data sources are unavailable.

```
## Final panel rows: 30 across 3 clusters
```




```
##
## Final therapeutic panel: top 10 candidates per cluster:

## # A tibble: 30 x 8
##    cluster gene    logFC     q mean_ess top_drug max_phase panel_score
##    <chr>   <chr>   <dbl> <dbl>    <dbl> <chr>        <dbl>       <dbl>
##  1 C1      PCNA    1.19      0   -2.96  <NA>            NA        4.24
##  2 C1      PUF60   0.744     0   -2.92  <NA>            NA        3.18
##  3 C1      PRPF19  0.64      0   -2.70  <NA>            NA        2.62
##  4 C1      SF3A3   0.685     0   -2.22  <NA>            NA        1.96
##  5 C1      SF3A1   0.723     0   -2.00  <NA>            NA        1.72
##  6 C1      UBE2I   0.828     0   -1.73  <NA>            NA        1.53
##  7 C1      ACTL6A  0.763     0   -1.70  <NA>            NA        1.34
##  8 C1      RBM25   0.747     0   -1.61  <NA>            NA        1.16
##  9 C1      EFTUD2  0.671     0   -1.72  <NA>            NA        1.16
## 10 C1      SSRP1   0.899     0   -1.36  <NA>            NA        1.12
## 11 C2      NSF     0.988     0   -2.44  <NA>            NA        2.98
## 12 C2      PACSIN1 2.54      0    0.039 <NA>            NA        2.63
## 13 C2      NEFM    2.49      0    0.126 <NA>            NA        2.38
## 14 C2      CAMK2B  1.59      0    0.081 <NA>            NA        2.12
## 15 C2      NEFL    2.49      0    0.331 <NA>            NA        2.07
## 16 C2      STXBP1  1.96      0   -0.191 <NA>            NA        1.69
```

```
## 17 C2      VSNL1   2.10      0    0.035 <NA>            NA       1.65
## 18 C2      STX1B   2.00      0   -0.069 <NA>            NA       1.59
## 19 C2      PRKCG   1.77      0    0.113 <NA>            NA       1.55
## 20 C2      TUBB4A  1.85      0   -0.204 <NA>            NA       1.45
## 21 C3      SUSD5   2.16      0   -0.018 <NA>            NA       1.86
## 22 C3      CSPG4   1.98      0   -0.136 <NA>            NA       1.63
## 23 C3      IL1RAP  1.97      0    0.007 <NA>            NA       1.41
## 24 C3      ITGAV   1.34      0   -0.789 <NA>            NA       1.23
## 25 C3      RAB3IP  1.65      0   -0.009 <NA>            NA       0.719
## 26 C3      NLGN3   1.64      0   -0.002 <NA>            NA       0.686
## 27 C3      LRP1    1.49      0   -0.118 <NA>            NA       0.514
## 28 C3      RAB31   1.57      0    0.011 <NA>            NA       0.5
## 29 C3      HIP1    1.52      0   -0.027 <NA>            NA       0.449
## 30 C3      EPN2    1.52      0   -0.005 <NA>            NA       0.421
```

15. Molecular Surrogate Classifier

The §10–11 survival arm is descriptive only because the cohort has too few OS events to support a credible per-patient predictive model (rule of thumb: 10 events per parameter). This section turns the predictive question on its side: rather than predicting survival, we predict molecular outcomes that themselves drive survival and that today require sequencing, WHO grade and the canonical pediatric brain tumour driver flags. With 100+ patients per outcome class, the events-per-parameter ratio is comfortable and a properly held-out evaluation is honest.

The deliverable is a proteomic surrogate classifier: given a tumour's proteome, predict its grade or driver status. Such a classifier could in principle inform treatment decisions where mutational testing is unavailable, slow, or inconclusive.

15.1 Define classification targets

```
## Classifier cohort: 199 patients

## Class distributions:

##   Grade:

##
##  LowGrade HighGrade
##       133        66

##   Any driver:

##
##      None AnyDriver
##       180        19
```

15.2 Cross-validated logistic-lasso classifier per target

`cv.glmnet` with `family = "binomial"` and `keep = TRUE` returns the out-of-fold linear predictor for every patient, i.e. each patient is scored by a model that was never trained on them. This is the only honest way to assess a high-dimensional classifier on a single cohort.

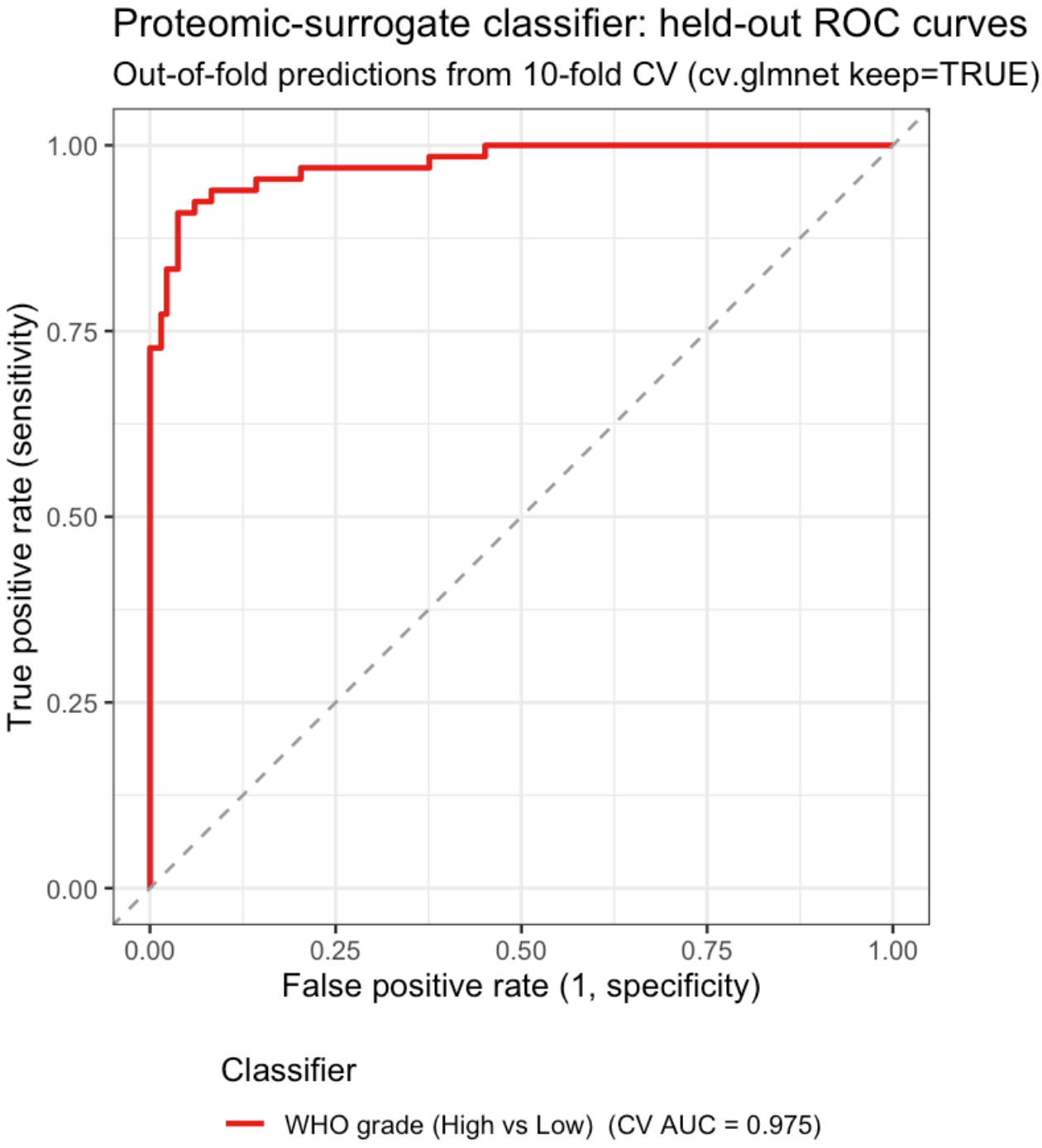
```
## WHO grade (High vs Low) | n=199 | events(HighGrade)=66 | CV AUC = 0.975 ± 0.010 | selected proteins =

## Any known driver mutation | n=199 | events(AnyDriver)=19 | CV AUC = 1.000 ± 0.000 | selected protein
```

Note

Only 19 of 199 patients have a known driver mutation. That's a 10:1 class imbalance with n=199. A cross-validated AUC of exactly 1.0 with SE=0.000 in this setting almost certainly means the proteome is partially

encoding histological subtype, and the driver flags are also subtype-correlated (e.g. BRAF mutations are concentrated in LGG, which also has a distinctive proteome). The classifier is likely learning subtype, not driver status directly.

15.3 ROC curves with held-out predictions



15.4 Selected proteins per target

```
##
## WHO grade (High vs Low), 22 proteins selected at lambda.min:
```

```
##  [1] "CASP3"   "EHD3"    "EPB41L2" "EPS8L2"  "GGH"     "HUWE1"   "IPO9"
##  [8] "JMJD6"   "LUC7L"   "OPLAH"   "PDLIM2"  "PSMD2"   "SGTA"    "SMC2"
## [15] "SORBS3"  "TNS3"    "TRMT6"   "TRMT61A" "UBE2A"   "WDR81"   "XPO5"
## [22] "YWHAQ"
##
## Any known driver mutation, 10 proteins selected at lambda.min:
##  [1] "ISOC1"   "KRT14"   "PACSIN3" "PLCB3"   "RPS27A"  "TRIP6"   "ZNF185"
##  [8] "DUSP14"  "KRT85"   "RASSF4"
```

15.5 Limitations

- **Single-cohort.** CV AUC is the most defensible internal estimate, but external validation in a second cohort (PBTA, OpenPedCan, an adult CPTAC release) would be the proper next step before any clinical claim.
- **Driver-flag definition.** `any_known_driver` collapses six biologically distinct events into one binary label. A multinomial model that predicts driver *type* (BRAF vs H3F3A vs CTNNB1 vs ependymoma fusion) would be the natural extension once cohort size allows.
- **What this classifier is and is not.** A high-AUC grade classifier is a *biomarker surrogate*, not a treatment-response predictor. The clinical use case is "saves a sequencing test," not "tells you what therapy will work." Therapy selection is what §14 is for.

References

- Petralia F. et al. (2020) "Integrated Proteogenomic Characterization across Major Histological Types of Pediatric Brain Cancer." **Cell** 183(7), 1962–1985.e31. doi:10.1016/j.cell.2020.10.044
- Hofree M. et al. (2013) "Network-based stratification of tumour mutations." **Nature Methods** 10, 1108–1115.
- Monti S. et al. (2003) *Consensus Clustering.* **Machine Learning** 52, 91–118.
- Cerami E. et al. (2012) *The cBio Cancer Genomics Portal.* **Cancer Discovery** 2, 401–404.
- Ishwaran H. et al. (2008) "Random survival forests." **Annals of Applied Statistics** 2, 841–860.
- Casado P. et al. (2013) "Kinase-substrate enrichment analysis provides insights into the heterogeneity of signaling pathway activation in leukemia cells." **Science Signaling** 6(268), rs6. doi:10.1126/scisignal.2003573 — *KSEA methodology used in §14.2.*
- Türei D. et al. (2016) "OmniPath: guidelines and gateway for literature-curated signaling pathway resources." **Nature Methods** 13, 966–967. — *Kinase–substrate annotations used in §14.2.*
- Tsherniak A. et al. (2017) "Defining a Cancer Dependency Map." **Cell** 170(3), 564–576.e16. doi:10.1016/j.cell.2017.06.010 — *DepMap CRISPR essentiality used in §14.4.*
- Ochoa D. et al. (2023) "The next-generation Open Targets Platform: reimagined, redesigned, rebuilt." **Nucleic Acids Research** 51(D1), D1353–D1359. — *Druggability evidence used in §14.5.*
- Friedman J., Hastie T., Tibshirani R. (2010) "Regularization paths for generalized linear models via coordinate descent." **Journal of Statistical Software** 33(1), 1–22. — *`glmnet` cross-validated lasso classifier used in §15.*